

COMPILING THE FLY BRAIN



What a Wiring Diagram Constrains,
and What Experiments Must Decide

THE FLY-BRAIN PROJECT

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Revised research edition, September 2026

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Chapter 1

What Does a Wiring Diagram Actually Compute?

Imagine being handed a map of every road in a country. The map is extraordinarily detailed. It includes the narrow lane behind a house, the motorway connecting two cities, and the junction where several routes meet. You can answer questions that were previously impossible to ask. You can find bottlenecks, identify isolated regions, and work out where a new bridge would make the largest difference. But the map does not tell you where people will drive tomorrow morning. For that you need to know something about the people, their destinations, the traffic rules, and the conditions on the road.

A connectome creates a similar opportunity in neuroscience. It is a reconstruction of neurons and their connections, detailed enough to let us follow pathways through a nervous system. The male fruit-fly dataset used in this project contains 165,122 neurons in its packed representation. After the project's connection threshold is applied, the graph contains about 10.5 million connections accounting for 104 million reconstructed synaptic contacts. These are measurements of a particular representation of a particular animal, rather than a universal specification of a fly.

The immediate temptation is to load the graph into a simulator and press play. That is a reasonable experiment. We will do it. But there is a question to settle before deciding what its output means: how much of the resulting behaviour came from the wiring, and how much came from the choices required to make the wiring executable?

A synapse count has to become an electrical effect. A neuron has to acquire a threshold, a resting potential, and a rule for integrating its inputs. Some neurons communicate through spikes; others carry graded signals. A body has to

turn neural activity into forces, and sensors have to turn light, contact, and chemical concentration back into neural activity. Each choice supplies information. Some of that information comes from experiments. Some comes from mathematical convenience. Some remains an educated guess.

This book follows an attempt to make those choices visible. Its central object is the space of candidate computations that survives after we account for what the wiring actually constrains.

From a map to a hypothesis

Consider a group of neurons arranged around a circle. Nearby neurons excite one another, while a second population suppresses activity on the opposite side. It is natural to suspect that the circuit can maintain a localised patch of activity. The position of that patch could represent an angle: which direction the animal faces, for example.

Already we have crossed several boundaries. The arrangement around a circle is an anatomical observation. The excitation and inhibition depend partly on transmitter assignments and receptor assumptions. The stable patch is a dynamical hypothesis. The interpretation of its position as heading is a hypothesis about representation. These statements support one another, but they are different statements.

To see why the distinction matters, keep the connections fixed and lower the excitatory gain. The patch may disappear as soon as the input stops. Raise the gain too far and activity may spread around the whole circle. Change the resting excitability and the same graph may alternate between silence and a persistent patch. None of these outcomes changes which neurons are connected.

The wiring has narrowed the possibilities. It has made a heading memory plausible and supplied a geometric explanation for it. It has not selected an operating point. A useful analysis should therefore return more than an attractive simulation. It should describe where the candidate computation appears, which assumptions it needs, and what could distinguish it from a different computation using the same connections.

That is the motivation for treating connectomics as a problem of constrained inference. We start with anatomy, propose a family of dynamical models, and ask what remains true as the unmeasured quantities vary. When the models disagree, we ask which intervention would make that disagreement observable.

Why the compiler analogy helps

A compiler turns a program written in one language into something that can run in another. It usually does this through intermediate representations: simplified descriptions that make particular properties easier to analyse. A compiler can check types before allocating registers because the two tasks need different views of the same program.

The project uses a related organisation. It first turns the connectome into a common graph representation. It measures recurrence, inhibition, convergence, and spatial organisation. It interprets combinations of those measurements as candidate computational operators. Each supported operator can then be connected to a specification of model parameters, stimuli, observables, and perturbations. An execution stage runs the resulting ensemble and records how its members differ.

There is a limit to the analogy. An ordinary compiler begins with a program whose semantics are already defined. A connectome does not arrive with such a definition. No language specification tells us that a particular reciprocal loop must store a memory or compare two choices. The inferred computation is part of what we are trying to learn.

It is therefore helpful to think of the system as a decompiler that produces hypotheses. It tries to recover candidate computations from the physical organisation of a machine, while retaining the uncertainty that ordinary decompilation often hides. The product is a set of executable possibilities and a way to investigate them.

This organisation also makes errors easier to locate. If an analysis reports a circular structure, we should be able to inspect the positional mapping that produced the circle. If a model uses inhibitory connections, we should be able to recover the sign convention. If an experiment is ranked highly, we should be able to see the particular members and outcome categories responsible for its score. The chain of reasoning should survive inspection at every stage.

What counts as agreement

Suppose two models follow the same moving landmark. Their decoded heading traces overlap almost perfectly. It would be easy to treat them as equivalent explanations of the circuit.

Now suppose one maintains its estimate through strong local recurrence, while the other is continuously refreshed by external input. Remove the landmark and the models separate. Or suppose both remember heading in darkness, but

one depends on an inhibitory population that the other can lose without difficulty. Silencing that population separates them even when the original task does not.

Agreement on behaviour is therefore only one kind of agreement. Neural activity can reveal differences that a behavioural score hides, and perturbations can reveal differences that ordinary activity hides. A convincing explanation should make contact with all of them, at the level of detail relevant to its claim.

This does not mean every model must reproduce every spike. A model of heading estimation may reasonably compress thousands of spikes into one angle. But the compression should preserve the distinctions we care about. If two mechanisms make different predictions about bump width after inhibition is reduced, a single heading-error score cannot settle the dispute.

The ensemble approach uses these differences constructively. Rather than asking which parameter setting makes the most appealing demonstration, it asks which candidate experiment divides the sampled models into different outcomes. The first implementation uses a simple, unweighted pair-separation score. That is a starting point for experimental design, with important limitations that we will examine. It is not yet a calculation of the best experiment an animal laboratory should perform.

The three circuits

The investigation begins with the heading circuit in the central complex. This is a useful starting point because anatomy and physiology have already given it a substantial interpretation. A localised activity bump tracks heading, and several named populations participate in stabilising and updating it. The project can ask whether its structural analysis recovers the relevant organisation and whether its models reproduce particular responses.

A known circuit is an unusually demanding test if we resist putting the answer into the detector. It lets us separate rediscovering an anatomical pattern from predicting a physiological outcome. It also exposes the temptation to use familiar biology as background knowledge and then quietly count agreement with that same biology as validation.

The second circuit connects PFN neurons to $h\Delta B$ neurons in the fan-shaped body. Here the candidate computation is a transformation between spatial representations. The wiring contains systematic column offsets, which suggest a shifted output. The experiment asks whether the dynamical response behaves like a fixed shift and whether recurrent pathways are required to produce it.

The third circuit is the mushroom body. Its expansion into Kenyon cells, extensive inhibitory feedback, and compartmental output suggest a family of memory-related computations. The implemented experiment is narrower: it probes response separation and gain under direct Kenyon-cell stimulation. It does not exercise the full sensory expansion, and it does not train and retrieve an association. Recognising that boundary turns out to be essential to interpreting its negative result.

Together these cases test different kinds of observable. Heading lives on a circle. The PFN experiment uses a column coordinate. The mushroom-body probe compares patterns of activity without requiring a spatial axis. This variety makes it harder for the machinery to succeed merely because every question has been made to resemble the first one.

The body as an integration test

A second branch of the project connects the neural simulation to a fly body with physical joints, muscles, and senses. This branch is visually compelling, but its scientific role is easiest to understand as an integration test.

A circuit can behave sensibly under an isolated stimulus and become unstable when its output changes its own input. A moving foot produces touch. A turn produces optic flow. A grooming leg can enter the visual field. Once those loops close, a model must distinguish an external event from a consequence of its own action, or tolerate the consequences of failing to distinguish them.

The embodied model also makes supplied behaviour unusually visible. If a stepping generator sits between descending neurons and the legs, walking in the animation is evidence about a combined system. If a separate process schedules bouts of activity, spontaneous movement does not show that the neural graph generated the bout statistics. Those additions may be useful and well grounded. Their location in the causal chain determines what the demonstration establishes.

We will therefore follow the signal far enough to ask where a decision, a rhythm, or a correction entered. The relevant boundary is between a component's observed contribution and the larger behaviour that depends on several components together.

Reading the investigation

The foundations chapters introduce the dataset, its intermediate representation, and the structural patterns that motivate the candidate operators. The

method chapters explain how an operator becomes a sampled model family and how perturbations expose differences within it. The circuit chapters then work through the consequences, including cases where the first interpretation has to change.

The assessment chapters ask whether a reduced heading estimator is useful as an engineering algorithm, how much of the pipeline is actually generic, and what happens when the model is embodied. The appendices explain how to trace results to their computational and scientific sources without interrupting the main argument with implementation instructions.

A recurring observation will be that a compelling structural signature does not guarantee the proposed computation in the models tested. The earlier draft described this as structure overstating what dynamics delivers. That phrase captures the experience of the investigation, but it needs a careful reading. The dynamics are hypotheses too. Failure can reveal an unmeasured parameter, an unsuitable neuron model, a missing input, or a poorly chosen probe. It does not by itself reveal which explanation is correct.

The useful outcome is a more specific question. Why did the bump disappear? Which recurrent pathway supported the shift? Would the inhibitory neuron behave differently if its activity were graded and local? What observation would separate these explanations? A connectome becomes scientifically more valuable when it helps us ask such questions precisely, even when it cannot yet answer them.

Chapter 2

Reading the Connectome

The first modelling decision happens before a neuron receives a time constant. It happens when we decide what counts as a neuron, a connection, and an observation worth retaining. A graph can look like raw data while already incorporating thresholds, aggregation, and interpretation. Understanding that representation is the beginning of understanding every later result.

The project uses the male central nervous system reconstruction known as MaleCNS, version 1.0. Its packed graph contains 165,122 neurons assigned to 11,752 cell types. Connections with fewer than three reconstructed synapses have been omitted. The retained graph has 10,511,038 directed connections, representing 104,213,652 synaptic contacts. A connection can contain as few as three contacts or as many as 2,591.

These quantities describe different objects. If one neuron makes twenty contacts onto another, that contributes twenty synapses but one directed connection. If the second neuron also contacts the first, there are two directed connections. If both neurons belong to the same named type, a type-level graph may fold all of this into a single population relation. Confusing these levels can change a ratio by orders of magnitude while leaving the sentence around it apparently plausible.

The graph in this book is a thresholded derivative of the reconstruction. Its counts should not be read as totals for every synaptic contact in the original release. Later analyses sometimes impose stricter thresholds of five or six contacts, so even within the project there is no single graph size that describes every calculation. The threshold belongs with the result.

Counting is not measuring strength

Imagine two connections with twenty contacts each. One lands near a region of membrane that strongly influences spike generation. The other lands on a distant branch. The postsynaptic cells may differ in size, resistance, receptor expression, and recent activity. Even if the contacts look equally convincing in the reconstruction, their electrical effects need not be equal.

The count is still valuable. Twenty observed contacts generally provides different anatomical evidence from one uncertain contact. Across a population, contact counts constrain where a model should place its interactions. But turning a count into conductance requires an additional rule. A global multiplier is one such rule. A multiplier adjusted for cell size is another. Neither becomes an observed physical quantity merely because it is applied consistently.

This distinction is especially important when a model fails. If a feedback loop is too weak, increasing its conductance may repair the result. That shows that the model can operate at a different gain. It does not establish that conductance was the only missing ingredient. The representation of the inhibitory cell itself may have discarded local dynamics, and the test stimulus may have driven the circuit outside the range in which the loop normally operates.

The project also measured how well connection counts compress. Encoding counts alone required about 4.04 bits per count; the best tested contextual encoding required about 3.95. The small difference says that those particular context models found little additional predictive information. It does not prove that counts are statistically independent of the rest of the graph. A better model might exploit a relation that these compressors missed.

There is a broader distinction here between a successful prediction and the absence of one. Finding a pattern is evidence that the pattern exists in the tested data. Failing to compress a variable further is evidence about the predictor as well as the variable. We will encounter the same logical asymmetry when a simulator fails to produce a computation.

Labels carry assumptions and geometry

Each neuron is accompanied by annotations such as type, hemisphere, superclass, transmitter prediction, and volume. These labels make the graph tractable. They let us select all Kenyon cells, distinguish descending neurons from local interneurons, and compare homologous populations across the two sides of the animal.

Some labels contain spatial information. Instance names in the protocerebral bridge encode side and glomerular position. Names in the fan-shaped body can encode a column. Olfactory receptor-neuron names identify glomerular channels. A graph that appears to have no coordinates can therefore contain an implicit coordinate system in its annotations.

This is useful, but it changes what a discovery means. Recovering a ring from anatomically assigned column labels is a finding about connectivity organised by known position. It is not the same as discovering a circular latent space from an unlabelled adjacency matrix. Both are worthwhile tasks. Their inputs differ, and the claim should acknowledge the difference.

The same caution applies to type definitions. A fine type can preserve a distinction that disappears when several populations are merged. Conversely, an analysis at individual-neuron resolution can be dominated by repeated copies of a common cell type. There is no universally correct level of aggregation. The appropriate representation depends on whether the question concerns individual connections, repeated local circuits, or interactions between populations.

Positional labels also require a convention. Eight analysis bins around a full circle correspond to 45 degrees per bin. Sixteen anatomical wedges correspond to 22.5 degrees per wedge. A bridge coordinate is not automatically either one. The hemisphere mapping and the folding operation must be specified before converting an offset into an angle. In this book, measured offsets remain in their stated analysis coordinates unless an explicit conversion is justified.

What a sign means in this project

A structural analysis often needs to distinguish excitatory from inhibitory connections. The project uses transmitter predictions to construct a sign proxy, but the proxy is not identical across all stages.

The stored fly sign representation treats acetylcholine as positive and GABA, glutamate, and histamine as negative. It also inherits positive contributions for monoamines in the convention used by the packed fly analysis. For neurons without a consensus transmitter assignment, the graded proxy combines the predicted probabilities: acetylcholine and monoamines contribute positively, while GABA, glutamate, and histamine contribute negatively. There are 3,602 neurons for which this graded treatment matters.

This explains why some dopamine-associated pathways appear among positive-positive motifs in the structural report. Such a motif is positive under the report's proxy. It is not proof that both pathways act as fast excitatory synapses

in the animal. Transmitter identity alone does not determine receptor-specific effects or timescale.

The worm loader uses a different convention: acetylcholine is positive, GABA and glutamate are negative, and monoamines are zero. The embodied fly model introduces another distinction by treating octopamine as a slow modulatory influence rather than retaining its ordinary fast contribution. The circuit ensemble builder, meanwhile, uses the neuron model's transmitter table unless explicitly overridden. It does not automatically inherit every sign or parameter choice used by the embodied model.

These differences are part of the current method. A shared graph interface makes them visible; it does not make them disappear. Sign-conditioned comparisons must carry their conventions, and a future comparison should rerun both datasets under harmonised alternatives. In particular, glutamate's effect is a modelling assumption that can depend on receptors, rather than a universal inhibitory label.

Unknown and zero also need care. A neuron without a consensus call may receive a nonzero graded sign in one representation. A truly zero-sign connection contributes no fast effect in a particular simulation. Omitting an effect avoids inventing a direction, but it still changes the circuit. Silence caused by omitted inputs is not evidence that the biological inputs do nothing.

A common representation

The intermediate representation keeps the information needed by generic analyses in a small number of conceptual groups. It records each cell's identity and classification, its side, its sign proxy, and whether it is treated as a sensory or motor boundary. It stores chemical connections as directed adjacency lists with contact counts. When electrical connections are available, it stores them separately as symmetric relations. Provenance accompanies these structures so that thresholds and conventions remain recoverable.

The chemical graph is represented sparsely. For each source neuron, the representation gives a contiguous list of target neurons and corresponding counts. It also records where each source's list begins and ends. This avoids allocating a square matrix with an entry for every possible pair of cells. Most possible pairs are absent, so a dense matrix would spend almost all its space recording zeros.

The significance of this choice goes beyond memory use. A shared representation means the flow calculation does not need to know which institution released the dataset or how its tables were arranged. The motif calculation sees a

graph and an explicit sign convention. The loader is responsible for translating the source, and the analysis is responsible for interpreting the translated object.

That separation creates a useful check. The fly loaded through the common representation should reproduce the earlier fly-specific structural calculations when the thresholds and conventions match. If it does not, the disagreement is first a data-translation problem. There is little value in interpreting a cross-species difference until the same-species translation has been checked.

Looking at another nervous system

The second loader reads the adult hermaphrodite worm connectome from Cook and colleagues' 2019 reconstruction as distributed through Netzscheuler. The resulting graph includes 454 cells, rather than only the 302 neurons conventionally associated with the animal, because the represented network also includes other cellular endpoints. It contains 4,879 chemical connections and 28,113 contacts under the project's processing choices. Bilateral homologues and serially repeated neurons are grouped into 169 classes.

Unlike the fly representation, the worm data supplies an electrical graph. Gap junctions can therefore remain a distinct edge type instead of being guessed from chemical wiring. That matters because a chemical synapse and an electrical connection do not have interchangeable dynamics. Combining them into an undifferentiated weight would simplify the interface by erasing a real biological distinction.

The common analyses produce striking contrasts. Under the chosen harmonic ordering, about 72 percent of worm chemical weight is classified as feedback, compared with about 12 percent in the fly. The fly instead has about half its weight classified as lateral. About 61 percent of worm cells belong to its largest strongly connected component, compared with about 97 percent of fly neurons. Reciprocal weight is closer: approximately 29 percent and 27 percent respectively.

These are descriptions of the processed graphs under a particular ordering. It would be premature to call one animal a feedback machine and the other a feedforward machine. Sensory and motor boundary annotations, cell inclusion, thresholds, and unknown transmitter assignments differ. The comparison shows that the same measurement can expose different organisation. Explaining how much of the contrast belongs to biology requires controlling the representation choices.

Transmitter coverage is a particularly large limitation. About 62 percent of worm weight is unsigned under the selected table, compared with roughly half a

percent of the fly's stored weight. A sign-conditioned worm motif census therefore observes a much less complete effective graph. Missing signs are not a small correction that can be ignored after quoting the counts.

What survives compression

A different kind of compression reveals useful regularity in the fly. Renumbering neurons by type and then by the spatial position of their skeleton centroids reduced the estimated cost of target identities from about 11.8 to 8.2 bits per connection. Around 28 percent of a neuron's targets also appeared among the preceding neuron's targets in this ordering.

Here the positive result is direct: type and position help predict connectivity in the tested encoding. Neighbouring, similarly classified cells have overlapping target repertoires. That regularity motivates population-level analyses and repeated-kernel interpretations, while leaving room for individual exceptions.

The representation still omits important physical information. It does not supply a full receptor map, cell-specific membrane dynamics, or all electrical coupling. It drops weak connections at packing time. A skeleton centroid also compresses an extended, branching cell into one location. These choices make the system analysable, and each one limits a different class of inference.

By the end of this stage we have a graph whose meaning can be stated precisely. We know what its weights count, how its signs were assigned, which coordinates came from annotations, and which source distinctions survived translation. That is enough to ask substantial structural questions. It is also enough to recognise why a structural answer cannot silently become a complete dynamical explanation.

Chapter 3

Structure Before Dynamics

A nervous system can be examined without simulating it. Paths reveal where influence could travel. Reciprocal connections reveal where feedback could occur. Population structure reveals repeated patterns that might implement similar operations in different places. These observations do not require us to choose a membrane time constant, and that independence is useful: they provide constraints against which later dynamical models can be checked.

The difficulty is that graph measurements are often easier to compute than to interpret. A short path is not a processing time. A large eigenvalue is not an observed instability. A frequent motif is not an algorithm. This chapter develops the measurements alongside the distinctions needed to use them.

Short routes through a recurrent network

Start at every annotated sensory neuron and follow connections containing at least five synapses. For each cell, record the shortest number of steps from any sensory source. About 15,900 cells begin at the sensory boundary. Another 27,500 are one step away, about 84,700 are two steps away, and about 35,200 are three steps away. Only a small remainder is farther away or unreachable under this threshold.

Descending and motor populations also have short available routes from sensation. Roughly half the descending neurons are one step from a sensory neuron, and most of the rest are two steps away. Of 708 motor neurons, 465 have a direct sensory connection in this graph.

These numbers show widespread opportunities for rapid access. They do not bound the depth of the computation that a cell participates in. Consider a motor neuron with both a direct touch input and a long pathway through several

interneurons. Its shortest sensory distance is one, even if the long pathway determines most of its behaviour. Adding a direct connection can collapse a shortest-path statistic without removing any of the deeper computation.

The same applies to time. A direct but weak connection may have less influence than a longer, stronger pathway. Recurrent activity can keep information in circulation, but long feedforward routes can also coexist with short ones. The graph therefore supports the conclusion that short sensorimotor routes are common. It does not support the stronger conclusion that all extended processing must occur through recurrence.

To describe the network more continuously, the project assigns a harmonic depth. Sensory cells are fixed at one boundary and motor or efferent cells at the other. Intermediate cells receive positions determined by an averaging relation over their partners. This produces an ordering in which antennal-lobe local neurons are close to sensory inputs, projection neurons and Kenyon cells lie somewhat farther along, and descending and nerve-cord populations lie nearer the motor boundary.

Relative to this ordering, approximately half the fly's synaptic weight is lateral, 37 percent is forward, and 12 percent is feedback. The rounded shares do not sum exactly to one. More importantly, the words forward and feedback refer to this inferred ordering. They do not identify a universal sequence through which every signal must pass.

What it means to belong to one loop

A strongly connected component is a set of nodes for which every node can reach every other by a directed path. About 160,500 neurons, or 97.2 percent of the packed fly graph, belong to one such component. This includes the central populations through which sensory interpretation and motor control interact.

The result rules out a picture of the central nervous system as a collection of wholly isolated boxes. There are routes connecting the boxes in both directions. But strong connectivity is a topological property. It says nothing by itself about whether an influence is strong enough, fast enough, or active in the relevant state to matter physiologically.

A useful analogy is a railway network. If every station can eventually reach every other, the network is strongly connected. That does not imply that a train from one station arrives at every destination within an hour, or that every route carries comparable traffic. The distinction is especially important in a neural graph whose paths can include inhibitory and modulatory interactions.

Reciprocity gives a more local view. About 19 percent of directed connections have a connection in the reverse direction, and these reciprocal connections account for about 27 percent of retained synaptic weight. Mutual influence is therefore common even before considering longer loops.

Under the stored fly sign convention, positive-negative reciprocal pairs are the largest category: about 616,000 neuron pairs, compared with roughly 264,000 positive-positive and 126,000 negative-negative pairs. These counts motivate a serious role for feedback inhibition. They do not establish that every positive-negative pair implements stabilisation; delays, operating points, and indirect pathways can change the outcome.

Inhibition and the left-right problem

The structural sign proxy assigns about 61 percent of synaptic weight to the positive side and 38 percent to the negative side. The median neuron receives around 44 percent of its input weight from negatively classified sources, with considerable variation across cells and regions. Central-complex populations receive a particularly large inhibitory share, while Kenyon cells and mushroom-body output neurons have smaller shares by count.

A proportion of inhibitory contacts is not the same as a proportion of inhibitory current. Nevertheless, it makes clear that inhibition is a substantial part of the architecture. A model that treats it as a small correction to excitation has already made a strong assumption about relative efficacy.

Connections crossing the midline account for about 22 percent of weight across the full graph. The proportion is low in the optic lobe and much higher among ascending, descending, and nerve-cord populations. Crossing connections are also more often assigned an inhibitory sign than connections that stay on one side.

This organisation suggests several possible uses. Inhibition can coordinate opposite sides, prevent conflicting actions, or sharpen the difference between competing sensory estimates. The strongest mutual inhibitory relations between homologous types include ring-neuron populations associated with the central complex. Yet an inhibitory bridge does not tell us whether the two sides alternate, select a winner, settle into a compromise, or remain inactive. Those are alternatives for a dynamical test.

Bilateral comparison is valuable for another reason. A reconstruction of one animal need not be exactly symmetric. Some differences may be biological; others may involve reconstruction or annotation. A model can amplify a small asym-

metry into a persistent turn. Before interpreting such a turn as a neural strategy, we should inspect the symmetry of its inputs and the sensitivity of the readout.

From recurring patterns to motifs

To count population motifs, the project aggregates the graph into its 11,752 types. It retains a type-to-type relation when the total contact count and the count per target neuron both exceed specified thresholds. Under the chosen rule, the resulting graph has about 473,500 directed type relations.

One common pattern is feedforward inhibition. An excitatory source reaches a target directly and also reaches an inhibitory population that projects to the same target. In the fly report, about 80 percent of strong positively classified type relations have such a parallel inhibitory route.

The motif is compatible with several computations. If excitation arrives first and inhibition follows, the target may respond mainly to changes or brief coincidences. If inhibition scales with broad population activity, the same wiring can contribute to gain control. If the inhibitory population has a different sensory tuning, it may implement competition between features. The graph identifies a route through which these effects could arise; it does not select the timing or tuning.

Another pattern is reciprocal inhibition. Two populations suppress one another. With sufficient gain and appropriate self-support, such a circuit can favour one state over the other. With different delays or adaptation it may alternate. With weaker coupling both populations may remain active. Calling it a winner-take-all candidate is reasonable as long as the word candidate survives into the interpretation.

The analysis also finds inhibitory populations whose main input group is also their main output group. APL, associated with the mushroom body, is a prominent example. Antennal-lobe local neurons and Delta7 in the central complex show related population feedback. This is a useful structural signature for regulation, but broad coverage does not prove that a neuron computes one global population total. Spatially local processing within an extended cell can matter even when the cell-level graph has collapsed it into a single node.

Feedforward loops, in which an indirect path accompanies a direct one, are also abundant. An excitatory intermediate can reinforce or delay a response; an inhibitory intermediate can truncate or oppose it. The same three-node drawing can produce different temporal computations when its synaptic and cellular dynamics change. For that reason, the catalogue records possible semantics separately from the measured motif.

Is the organisation surprising?

Large populations make many motifs almost inevitable. If one type connects to thousands of targets, it will participate in many triangles even in a largely shuffled network. Raw motif counts therefore need a comparison that preserves obvious sources of combinatorial abundance.

The project's null model rewires the type graph while preserving in-degree, out-degree, and source-sign constraints. The identities of connected partners change, while each type retains its broad connectivity budget. This asks whether the observed pairing of populations contains organisation beyond those retained statistics.

The answer is substantial under this null. Feedforward inhibitory coverage falls from about 80 percent in the observed graph to about 25 percent in the rewired graphs. Positive-positive reciprocal pairs fall from 8,272 to a null mean of about 526. Negative-negative reciprocal pairs fall from 6,375 to about 282. Both coherent and incoherent feedforward-loop counts also fall sharply.

The reported standardised deviations are very large, often several hundred null standard deviations. They should be read as descriptive contrasts against six rewired samples, not as calibrated probabilities of a biological hypothesis. Six samples provide only a rough estimate of null variance, and preserving degree does not preserve spatial locality, developmental constraints, or every feature of type composition. A more demanding null could retain some of these features too.

There is also a useful control. The total number of inhibitory-to-inhibitory edges remains unchanged under the stated rewiring constraints. A statistic that the null is designed to conserve should remain conserved. This helps check the manipulation without turning every observed quantity into an apparent discovery.

The defensible conclusion is that the arrangement of partners contains substantial structure beyond the chosen degree constraints. That supports further investigation of the motifs. It does not make a particular proposed computation hundreds of standard deviations more likely.

What the spectrum can reveal

Treating the signed contact-count matrix as a linear operator offers another view. Its largest-magnitude modes identify patterns that are strongly coupled under the selected weights and signs. In the fly calculation, the spectral radius is about 3,771, and the leading eigenvector is concentrated on a small effective

population, with a participation ratio of about fifty cells. Much of its mass lies among antennal-lobe local neurons.

A participation ratio describes concentration. A vector spread evenly over many cells has a large effective support. A vector dominated by a few cells has a small one. The result therefore points to a local set of strongly interacting populations inside a much larger graph. It provides a candidate circuit to inspect without requiring an annotation that labels those cells as a dynamical centre.

The raw spectrum does not establish instability of the biological network. For a simple rate model, local stability depends on the Jacobian of the dynamics, which includes leak, gain, and the derivative of the input-output nonlinearity at the operating point. A large eigenvalue of an uncalibrated count matrix becomes relevant only through those factors. In a conductance-based spiking model, the relation is more complicated still.

The worm analysis places its leading mode among head-steering populations such as RMD, RIA, and SMD. That is an interesting convergence with known anatomy. It remains a statement about the signed representation and its incomplete transmitter coverage, rather than a direct measurement of the animal's dominant oscillation.

These structural passes leave us with a more organised set of questions. We know where inhibition, recurrence, and repeated spatial patterns are concentrated. We know which observations depend strongly on aggregation and sign conventions. The next task is to look at individual regions closely enough to propose computations whose successes and failures can be measured.

Chapter 4

A Journey Through the Circuits

A global graph statistic can tell us that recurrence is common without telling us why a particular loop exists. To move from organisation to computation, we need to follow signals through identifiable populations. This chapter takes that closer view. The aim is to understand why several algorithmic interpretations are tempting, and what additional evidence each interpretation would require.

The anatomical names can initially feel like an obstacle. It helps to treat them as addresses. An address lets us return to the same population when a hypothesis becomes an experiment. We need not memorise every label, but we do need to preserve the difference between a named group of cells and a function that we hope those cells perform.

Smell begins with channels

The antennal lobe receives olfactory input through glomeruli, local regions associated with particular receptor channels. In the project's census there are fifty glomeruli, 2,562 olfactory receptor neurons, 267 uniglomerular projection neurons, and about 420 local neurons. A typical glomerulus contains a few dozen receptor neurons and a handful of projection neurons.

The direct receptor-to-projection wiring is remarkably channel-specific. About 96.6 percent of its weight remains within the same glomerulus. Each projection neuron receives from a median of thirty-eight receptor neurons, and each receptor neuron reaches about four projection neurons. This looks like a system that pools repeated measurements while retaining channel identity.

Pooling is useful when several receptors carry related signals. A downstream cell can combine their activity instead of relying on one noisy input. But preserving a channel is not the same as preserving an unmodified signal. The an-

tenal lobe also has extensive local interactions that can change the gain and selectivity of the projection-neuron response.

Receptor neurons make about 537,000 contacts onto local neurons, compared with about 414,000 onto projection neurons. Local neurons often span multiple glomeruli: the median cell receives from eight and projects to fifteen, while broadly connected examples cover most of the fifty. They also return approximately 169,000 contacts to receptor terminals. Such connections provide an anatomical route for presynaptic regulation of sensory transmission.

The local network is itself densely interconnected. Local-to-local weight exceeds local-to-projection weight, and its transmitter assignments are mixed. Some local cells are GABAergic or glutamatergic under the project's inhibitory convention; others are cholinergic. The latter include populations with very large outputs onto projection neurons.

This makes the circuit more interesting than a bank of independent labelled lines. Odour identity enters through relatively separate channels, while a broad network can regulate those channels in relation to one another. Whether this produces useful normalisation or excessive mixing depends on the dynamics. In the embodied calibration, downstream odour specificity remained poor across the tested configurations. The broad cholinergic pathways are plausible contributors, but their contact counts alone do not identify the cause of the failure.

Two routes out of the antennal lobe

Projection neurons distribute information to both the lateral horn and the mushroom body. In the structural report, total weight to the lateral horn is about 1.036 times the weight to Kenyon cells. The two destinations therefore receive comparable anatomical investment under this measure.

The lateral horn contains about 2,028 neurons in 422 types. Most are classified as output cells, while the remainder are local interneurons. A typical output cell pools input from around six projection-neuron types, and its strongest projection-neuron type contributes about a quarter of that input. This suggests combinations of channels rather than a simple copy of one glomerulus.

The common description of the lateral horn as an innate pathway and the mushroom body as a learned pathway is useful orientation. It should not suggest two sealed systems. Mushroom-body output neurons contact lateral-horn populations, and the horn reaches descending pathways. Learned information can therefore influence circuitry associated with innate responses.

Imagine an odour that normally elicits approach but has recently predicted an unpleasant outcome. The anatomical convergence provides places where

a learned signal could modify the original response. The graph does not reveal the learned weight or the decision rule. It tells us where to look for the interaction.

Expansion into Kenyon cells

The mushroom body presents one of the most recognisable computational arrangements in the connectome. A smaller population of projection neurons supplies a much larger population of Kenyon cells, each of which samples a small subset of the available inputs. The resulting representation can distinguish combinations of sensory channels that were less separable in the input.

The denominator matters when describing this expansion. The broad census includes 686 projection neurons, giving 4,064 Kenyon cells an expansion ratio of about 5.9. Among the 282 projection neurons identified as reaching the calyx, the ratio is about 14.4. Relative to the 267 uniglomerular projection neurons, it is about 15.2. These are different population comparisons, not competing estimates of one number.

A typical Kenyon cell samples about five projection neurons, with most sampling between three and eight. Each calyx-projecting neuron reaches a median of about fifty-six Kenyon cells. The sampling is relatively heterogeneous: the mean absolute correlation between projection-neuron target patterns is about 0.023, compared with about 0.013 after shuffling. This supports a weakly structured expansion, rather than literal independent random wiring.

To understand why sparse sampling might help, consider two odours that activate largely overlapping input channels. A cell requiring a conjunction of several inputs may respond to one combination but not the other. A large population of such cells offers many opportunities for differences to become visible. Whether the actual responses are sparse and separated depends on thresholds, inhibition, and input statistics as well as on the sampling pattern.

Not every Kenyon cell belongs to the same olfactory route. The census contains 283 with no projection-neuron input under its selection, many belonging to subtypes associated with other sensory modalities. This is another reason to keep the selected population explicit. An apparent missing input can represent a different sensory pathway rather than an incomplete cell.

Feedback and teaching in the mushroom body

Kenyon cells have extensive reciprocal contact with the two APL neurons. The report records about 210,000 Kenyon-to-APL contacts and 196,000 in the reverse

direction, with complete Kenyon-cell coverage under the selected graph. This is a strong anatomical basis for feedback inhibition.

It is tempting to draw APL as a single box that sums all Kenyon-cell activity and returns one global inhibitory signal. That is a modelling choice. Broad innervation does not imply that activity mixes uniformly throughout an extended neuron. Chapter 11 examines why a point-neuron approximation is especially consequential here and why a failed gain-control test cannot be attributed to synaptic strength alone.

The output stage contains ninety-seven mushroom-body output neurons, each receiving from a median of 313 Kenyon cells. A Kenyon cell reaches a median of eleven output neurons, and the total Kenyon-to-output weight is about 439,000 contacts. These outputs form a distributed readout of the expanded activity pattern.

Dopaminergic neurons add another anatomical relation. Roughly 340 cells in the selected teaching populations contact Kenyon-cell axons and output compartments. About seventy percent of dopamine-to-output-neuron weight lands in the matching compartment. This organisation is consistent with compartment-specific modification of the readout, but the learning rule and the history of learned values do not arrive with the graph.

The distinction between architecture and an executed memory is straightforward. A memory experiment must change something through experience and then test whether later behaviour or neural output depends on that change. A graph containing candidate teaching pathways is evidence for where such plasticity could act. A probe of fixed weights, even in the correct anatomy, does not by itself test learning.

Outputs from different mushroom-body compartments later converge on shared targets, including descending populations. The compartmental organisation therefore supports several stages of interpretation: sensory expansion, activity regulation, plastic readout, and recombination into action-related signals. Collapsing them into one label such as associative memory can be convenient, provided each experimental claim identifies which stage was actually exercised.

A compass made of populations

The heading circuit uses populations associated with the ellipsoid body and protocerebral bridge. EPG neurons provide the principal measured heading representation. Delta7 neurons contribute an inhibitory pathway. PEG and PEN

populations participate in recurrent and shifted pathways. Anatomical labels allow the project to assign these cells positions on an analysis ring.

The two-hop EPG-to-Delta7-to-EPG pathway has a striking spatial profile. At the source position the structural measure is about 115 in the report's units; halfway around the eight-bin ring it reaches about 1,770, before falling again toward the starting position. A constant-minus-cosine fit explains approximately 99 percent of the profile's variation.

This is the sort of interaction one would choose when designing a ring attractor. Activity near one position receives comparatively little inhibition from that pathway, while activity far away receives more. Local excitation can then support a patch that suppresses competitors at distant positions. The wiring also contains same-column EPG recurrence and reciprocal EPG-PEG connections, supplying plausible routes for local reinforcement.

The cosine fit is a property of an aggregated path-weight profile. It is not a measurement of a postsynaptic current. In a probe of the calibrated whole-brain model, the realised inhibitory profile retains the broad orientation but has a much weaker cosine fit, around 0.56. That difference records a change between two representations. It could involve response nonlinearities, sampling, and the probe, as well as the particular weight calibration.

The PEN pathway adds spatial asymmetry. Left and right populations return to different positions relative to their input. In the chosen folded coordinate, typical offsets are approximately positive 1.47 and negative 1.45 bins. Pooling the sides would make the average look close to zero and hide the organisation relevant to directional updating.

These offsets must not be identified directly with a 22.5-degree anatomical wedge. The project uses an eight-bin coordinate for several analyses, and the relation between bridge positions, hemispheres, and ellipsoid-body wedges needs to be carried through explicitly. What the structural result securely shows is a pair of oppositely oriented pathways. A calibrated angular-velocity transformation requires more work.

Ring-neuron populations supplying the compass also show strong mutual inhibition and disinhibitory chains. This gives the heading system several possible sources of regulation. A model that assigns all inhibition to Delta7 is a simplification of that anatomy, and Chapter 9 shows why that simplification matters when Delta7 output is reduced.

Shifts in the fan-shaped body

Behind the compass are column-organised pathways associated with movement and steering. Connections between PFN and $h\Delta B$ populations have systematic offsets in the anatomical column labels. The structural histograms peak near negative three columns for one PFN arm and positive two for the other.

A first interpretation would be a fixed coordinate shift. Drive one input column and obtain a correspondingly displaced output. But a histogram of contacts cannot tell us whether the resulting response is a rigid translation, a position-dependent transformation, or a recurrent computation whose output happens to have a displaced centroid.

The fan-shaped-body graph contains the recurrent connections needed to make these alternatives substantive. PFN populations have self-recurrence, $h\Delta B$ cells have internal interactions, and $h\Delta B$ projects back to PFN. A static drawing focused on the feedforward projection can leave all three out while still looking anatomically convincing.

The circuit also has unequal population sizes and unequal inputs across its arms. Forty PFNd cells and twenty PFNv cells do not necessarily provide comparable drive merely because each arm has a clean peak in an offset histogram. Chapter 10 follows this asymmetry into the executable models and separates what was tested on the two arms.

Downstream populations such as PFL3 and PFL2 contact descending pathways associated with steering. This provides a route by which internal spatial representations could influence movement. It does not establish that every intermediate is performing an independently identifiable coordinate transform. The operational question is whether an input-output relation survives changes of position, amplitude, and recurrent support.

Repetition in vision

The optic lobe offers a different structural regularity: many copies of related local circuitry across visual space. Forty-seven types have at least six hundred instances. For some strong type pairs, nearly every source neuron connects, partner counts are relatively consistent, and total connection weight varies within a limited range.

This resembles a shared spatial kernel. A local operation is repeated across the eye so that a feature can be detected at many positions. Some pathways are close to one-to-one, while others connect across several columns. The correspondence with convolution is useful, although finite boundaries, variable weights,

and anatomical irregularities prevent it from being exact weight sharing in the engineering sense.

The lamina pathways also separate into channels. L1 strongly contacts populations such as Mi1 and Tm3, while L2 strongly contacts other populations including Tm1, Tm2, and Tm4. This block structure is associated with the ON and OFF organisation of visual processing.

Direction-sensitive T4 and T5 subtypes provide another repeated pattern. The four T4 subtypes have very similar input compositions, with full input-vector cosine similarities around 0.93 to 0.95. Their spatial arrangements differ. Similar ingredients at different offsets are consistent with a bank of related directional filters.

Timing is indispensable to that interpretation. Two signals arriving from neighbouring columns do not reveal motion direction merely because both are present. Their relative timing and nonlinear combination determine which direction is preferred. The anatomy supplies candidate inputs and offsets; a motion model must also specify delays and dynamics.

Wide-field neurons then pool many local detectors. Horizontal-system populations receive from hundreds of selected T4 and T5 cells, while other lobula-plate populations pool different subtypes. This changes the scale of the representation, allowing local feature measurements to contribute to broad motion signals. Thousands of visual projection neurons subsequently distribute specialised outputs to the central brain, where visual information meets other senses and internal state.

From sensation to action

The descending system contains 1,314 neurons in the project's census. They receive a relatively small share of total brain output weight, yet each typically combines input from many types. The median descending neuron receives about 1,722 contacts from eighty-seven types.

Calling this a command bottleneck is suggestive, but the population includes different functions. Convergence could reflect action selection, sensory integration, state-dependent gain, or communication to specialised motor circuits. A narrow anatomical channel need not carry a single central decision.

Descending neurons also contact one another, and their output largely reaches nerve-cord interneurons rather than motor neurons directly. Only about six percent of descending output goes straight to motor cells. Ascending pathways provide substantial return input to descending populations, so the arrangement includes feedback from the body-related circuitry it influences.

The motor census contains 708 neurons. A typical motor neuron receives from many premotor partners, with a large inhibitory share and a strong contribution from nerve-cord interneurons. Motor neurons of the same type and side share premotor partners much more often than random pairs do. This is consistent with coordinated pools, although the timing required for posture and stepping is not given by shared input alone.

Escape makes the completeness problem especially concrete. The giant fibre receives convergent visual input, including LC4 and LPLC2 pathways, and is associated with rapid jump and takeoff responses. Yet a chemically reconstructed graph does not supply every relevant electrical connection. The embodied implementation adds a known giant-fibre-to-jump-motor pathway that was absent from the packed chemical representation. The resulting behaviour belongs to the augmented model.

State within the network

Small populations associated with circadian timing, sleep need, peptides, and monoamines interact extensively with fast sensorimotor circuits. In the selected state-cell census, the largest measured exchanges are with the central complex, with roughly 111,000 contacts in one direction and 101,000 in the other.

This organisation makes state-dependent behaviour plausible without placing state in a separate control centre. The compass can influence state-related populations, and those populations can influence the compass. Descending and mushroom-body outputs also receive state-associated contacts.

The difficulty is that a modulatory contact often needs information the graph does not carry. Which receptor is present? Does activation change threshold, adaptation, synaptic release, or another process? Over what timescale? Uniformly lowering the thresholds of all anatomical targets is one executable hypothesis, but it is a strong simplification.

The regional tour has now supplied a vocabulary of possibilities: channel pooling, population regulation, sparse expansion, persistent spatial state, directional updating, repeated visual filtering, and distributed motor control. Each possibility has an anatomical basis. The next chapter asks how to turn that basis into an explicit candidate operator without pretending that a familiar diagram has already established its computation.

Chapter 5

From Wiring Patterns to Candidate Computations

Recognising a familiar shape in a circuit is a powerful source of hypotheses. It is also a powerful source of mistakes. Once a diagram resembles a ring attractor or a random-projection memory, we begin to interpret ambiguous details in the language of that algorithm. The operator layer exists to make this transition explicit enough to inspect.

An operator is a proposed computation attached to a particular anatomical substrate. It names an input, a possible internal state, and an output. It also records the wiring measurements that motivated the proposal and the observations that would count against it. Its purpose is to connect structural analysis to an executable question.

For example, a ring operator takes directional sensory information and angular-velocity-related input, maintains a position on a circle, and exposes that position to a readout. The structural signature includes local recurrence, an inhibitory surround, and oppositely shifted pathways. The proposed semantics include persistence and directional updating. The signature and the semantics belong in the same entry, but they remain different kinds of evidence.

A vocabulary with provisional meanings

The project uses an instruction-set analogy for its catalogue. In a processor, an instruction has an agreed meaning: adding two registers produces a specified result. A biological operator has a weaker status. Its meaning is proposed from anatomy and must be checked against dynamics and experiments.

The catalogue contains twelve detections, representing nine broad candidate operations plus lower-confidence detections under alternative criteria. The strongest ring candidate is accompanied by candidates for reciprocal competition, spatial shifts, motion correlation, repeated visual filtering, sparse memory-related processing, feedforward inhibition, population normalisation, and descending convergence.

Keeping weaker detections is useful when their failure is visible. A candidate that nearly satisfies a ring signature can show whether the detector depends on a specific spatial relation or merely on generic recurrence. If the catalogue displayed only accepted examples, it would be harder to see how selective its definitions really are.

The vocabulary is intentionally smaller than the list of anatomical cell types. Thousands of types may participate in variations of a few computational arrangements. Conversely, one type can participate in more than one operation, depending on state and pathway. There is no requirement that the brain partition neatly into one population per instruction.

This is one reason to avoid treating the catalogue as a literal program recovered from tissue. Biological circuits share components and interact continuously. The operator view is useful when it preserves a relationship that can be tested, rather than when it forces every connection into an engineering analogy.

What the confidence scores do

The ring candidate receives a score of 0.70 in the existing catalogue. Reciprocal competition receives 0.60. Several visual, spatial-shift, and memory-related candidates receive 0.55, while the descending funnel receives 0.30. These numbers are hand-weighted combinations of evidence items.

They are not calibrated probabilities. A score of 0.70 does not mean that seventy percent of similarly scored anatomical patterns have been shown to be ring attractors. No such validation set has been established. Small differences between scores should therefore not determine a scientific conclusion.

The scores are better understood as a compact record of the detector's preferences. A highly specific spatial signature receives more support than generic convergence onto a small population. A realised inhibitory profile contributes differently from a raw contact-count pattern. The value lies in being able to inspect those contributions and disagree with them.

Even the phrase structural detection needs qualification here. Some catalogue entries incorporate dynamical checks from another stage, such as a realised Delta7 profile or a PEN push measurement. These entries combine anatomical

and simulated evidence. Their subsequent agreement with those same checks cannot be counted as an independent success.

A future catalogue could make confidence more meaningful by evaluating detectors on a collection of circuits whose functions were independently established. It could then ask how often a signature generalises and how sensitive it is to thresholds. Until that exists, the evidence record is more informative than the decimal score.

Making a prediction concrete

Take feedforward inhibition. The structural pattern says that a source reaches a target directly and through an inhibitory intermediary. A proposed temporal interpretation is that the target responds initially and is then suppressed. To test that interpretation, the model needs a stimulus with timing, an observable that measures temporal width, and an intervention that removes or weakens the inhibitory arm.

If the response broadens after that intervention, the outcome supports the proposed role under the tested conditions. If the response barely changes, several explanations remain: the arm may be weak, its timing may be wrong for this stimulus, another pathway may compensate, or the interpretation may be mistaken. An operator entry should make such alternatives easier to formulate rather than compress them into a binary success flag.

Reciprocal inhibition poses a related problem. A candidate competition circuit should be challenged with simultaneous drive to both populations, varying their relative strengths and initial conditions. A persistent winner, alternating activity, and a graded compromise are different outcomes. All can arise from superficially similar wiring. The experiment must measure the distinction that the operator claims to explain.

For a repeated visual kernel, a relevant question is whether a feature presented at different positions produces correspondingly translated responses. Similar input counts across columns motivate the question, but are not its answer. Weight variation, receptive-field boundaries, and nonlinearities can break transfer even when the average wiring looks repeated.

For mushroom-body processing, the prediction must be decomposed further. Sparse sensory expansion, activity regulation, associative learning, and downstream readout are separate capabilities. A direct Kenyon-cell probe of fixed output weights can examine only some of them. The broad memory label should never obscure that experimental scope.

Geometry is part of the hypothesis

A computation often becomes recognisable only after choosing the space in which its signals live. A heading angle wraps around: north just clockwise of a reference and north just counterclockwise of it are nearby. A linear centroid calculation can place their average on the opposite side of the circle if it ignores that geometry.

The PFN experiment uses a different coordinate: an ordered set of columns. This makes centroid offsets easy to compute, but introduces questions about boundaries, missing columns, and the relation of that linear coordinate to the underlying anatomy. A position-dependent shift can reflect real circuit organisation, the stimulation protocol, or the measurement convention.

The mushroom-body probe has no comparable positional axis. It treats activity as a vector across cells and asks whether two vectors become more or less similar. Choosing cosine similarity makes the observable relatively insensitive to overall amplitude, which is useful for pattern comparison but means gain must be measured separately.

Thus geometry is not just a plotting decision. It determines what counts as agreement, what information the readout discards, and which failures remain visible. An operator specification should carry its geometry as explicitly as its population names.

A worked transition to an ensemble

Suppose the detector finds a candidate ring. The obvious free quantities include local recurrent gain, inhibitory gain, shifted-pathway gain, and baseline excitability. The graph constrains where these quantities act, while the model family explores selected values.

The candidate's semantics suggest the probe. Initialise a localised activity pattern, remove the localising stimulus, and measure whether a coherent pattern remains. Then drive each directional arm and measure displacement. The counterfactuals suggest perturbations: weaken inhibition, remove the copy pathway, or reduce baseline excitation.

At this point the catalogue has become an executable investigation. The original detection did not establish that the circuit stores heading. It supplied a structured reason to ask a series of narrower questions. A persistent bump in some parameter settings would support compatibility. A failure in all sampled settings would motivate checking both the parameters and the model class. Neither outcome would enumerate every possible implementation of the anatomy.

The final result should therefore retain the route from substrate to test. Which spatial measurement motivated the ring? Which parameters were allowed to vary? Which background assumptions were held fixed? Which observation produced the outcome label? These are the details that turn a computational analogy into a scientific proposal.

What the catalogue leaves open

There are many operations a detector built around familiar motifs will miss. Some computations may be distributed across several regions. Others may depend primarily on timing, intracellular dynamics, or learning rather than a distinctive cell-level pattern. An undetected operator is not necessarily absent from the brain.

There is also a selection effect in choosing attractive examples. A ring with an excellent cosine fit is easier to describe than a heterogeneous circuit whose role is uncertain. The catalogue should be read as the set of candidates the current detectors know how to propose, not an inventory of everything the connectome computes.

Within that boundary, the approach is useful. It reduces a large anatomical graph to a collection of questions with named substrates and explicit measurements. The ensemble method then asks how those questions behave when the unmeasured quantities vary. That is where a diagram begins to encounter the consequences of being made executable.

Chapter 6

Exploring a Family of Models

Suppose we know exactly which cells connect in a candidate heading circuit, but do not know how strongly its recurrent synapses act. A single simulation forces us to choose a strength. An ensemble lets us keep several choices alive and observe what each implies.

This changes the object of the investigation. We no longer ask only whether one model produces a bump. We ask how bump formation depends on recurrent gain, whether a second inhibitory pathway changes the result, and which intervention separates models that otherwise look alike. The aim is to map part of the uncertainty left by anatomy.

The word part matters. A finite parameter grid is a sample from a chosen model class. It is not the complete set of programs permitted by the connectome. A family of point neurons excludes graded, compartmental, and many biochemical mechanisms before its first parameter is varied. Expanding the grid explores the chosen assumptions more thoroughly; it does not remove them.

Parameters and mechanisms

Some unknowns can be represented reasonably as numerical axes. The efficacy of a selected connection class can be multiplied by several gains. Baseline excitability can be raised or lowered. A synaptic delay can be varied over a range. These are parameter changes within a specified model.

Other unknowns change the model's structure. A cell may require several interacting compartments instead of one voltage variable. An inhibitory influence may arrive through a population omitted from the model. A modulatory pathway may change release probability rather than threshold. No amount of gain adjustment necessarily substitutes for those mechanisms.

The distinction is not absolute. Adding a second inhibitory pathway introduces new parameters, and setting its gain to zero can recover the simpler model. But it remains useful to ask whether a proposed explanation was represented at all. A model can be flexible in the wrong dimensions.

One way to see the danger is to imagine fitting a straight line to a curved relation. Testing many slopes and intercepts will find the best line. It will not establish that the underlying relation is linear. In neural modelling, the analogue of curvature may be local dendritic integration, receptor-specific modulation, or state-dependent input. A dense grid can give an impression of completeness while leaving the relevant possibility outside the family.

The actual neuron model

The circuit experiments use leaky integrate-and-fire neurons. Each cell has a voltage that relaxes toward rest while incoming activity changes its state. Crossing threshold produces a spike, followed by a reset and refractory interval. The model also includes adaptation and synaptic depression, so recent activity influences subsequent responsiveness.

The default circuit configuration uses a resting potential of negative 52 millivolts, a threshold of negative 45 millivolts, a membrane time constant of twenty milliseconds, and a synaptic time constant of five milliseconds. Integration proceeds in half-millisecond steps with a nominal synaptic delay of 1.8 milliseconds. These choices follow the project's LIF modelling lineage, rather than measurements of each cell in the male reconstruction.

The source path is important here. The ensemble builder loads the full packed network, applies cell-size scaling, and modifies selected edge classes and population biases. It does not extract only the named circuit cells. The circuit names identify the manipulated and measured substrate inside that full network.

The builder also does not automatically import the fitted parameters of the embodied model. Its default synaptic scale is 0.275, its size exponent is one, and conductance-based operation is disabled by default. Adaptation increases threshold by two millivolts per spike and relaxes over one hundred milliseconds; synaptic depression is also active. The embodied configuration described later uses different settings, including conductance-based synapses and zero spike-frequency adaptation.

This distinction resolves an apparent contradiction in the earlier description. An adaptation-related shutdown is possible in the circuit ensemble even though adaptation is zero in the embodied fit. The two experiments use different configurations. It also means that their agreement or disagreement cannot be in-

interpreted as a pure effect of stimulation while assuming every other parameter is shared.

The base configuration gives the ensemble its units and background dynamics. A recurrent gain of four multiplies the selected EPG-to-EPG weights by four relative to that base. It does not mean four times an experimentally measured synaptic conductance. The multiplier is explicit; the physical calibration remains a model assumption.

Constructing a useful sample

The heading ensemble varies recurrent excitation over four values, inhibitory gain over three, and tonic EPG bias over four, while holding the PEN-pathway multiplier fixed. That gives forty-eight members. The memory-related probe varies readout gain, APL gain, and output-cell bias to create eighteen members. The PFN experiment uses seventy-two combinations of pathway gains, recurrence, and bias.

A regular grid has practical advantages. It is easy to describe, rerun, and inspect. Nearby failures can reveal a regime boundary, and a reader can see which range was explored. But a grid also assigns weight to regions by construction. A parameter sampled at twice as many points contributes twice as many members to an unweighted comparison.

There is no reason to assume that biological plausibility is uniform in the chosen coordinates. Equal steps in conductance and equal steps in its logarithm describe different sampling distributions. A narrow region of stable operation may be biologically plausible if development regulates the relevant quantity tightly. Its small fraction of a broad grid is not a probability that the animal uses it.

The appropriate first claim is therefore descriptive: twelve of these forty-eight settings sustained a localised state under this probe. To turn that into a statement about likely biological mechanisms would require further constraints or an explicit prior over parameters and model classes.

Following one experiment through

Consider two ensemble members that retain a coherent bump after a localised stimulus ends. At baseline their decoded phases and widths are similar. In one, strong excitation is balanced mainly by Delta7 inhibition. In the other, a different background interaction supplies enough restraint that Delta7 contributes little under the probe.

Silencing Delta7 can expose this difference. The first member may spread into broad activity or lose its stable operating regime. The second may retain a concentrated bump. The perturbation is informative because the baseline observable left the alternatives unresolved.

The execution loop applies the same general sequence to all members. It builds a network with the chosen parameters, presents a defined stimulus, measures continuous observables, and assigns an outcome category. It then repeats the procedure under each perturbation. The report retains the parameter values, observations, and labels so that a coarse category can be traced back to its measurement.

The classification thresholds are part of the method. A circular concentration just above one half and one just below it may produce different labels despite very similar activity. Such boundaries are useful for summarising many runs, but a change of category should not automatically be called a change of mechanism. The continuous measurements need to remain available.

Randomness also requires discipline. A fixed seed makes a particular execution reproducible. It does not necessarily give two different models identical random inputs if their execution paths consume random values in a different order. The current implementation seeds a shared generator; stronger comparisons would provide matched stimulus streams explicitly for each member and perturbation. Reseeding and examining ranking sensitivity are useful, but they answer a different question from exact replay.

Understanding the separation score

The initial ranking uses unweighted ensemble separation. For an experiment, count how many pairs of different members receive different outcome labels, then divide by the total number of member pairs.

If there are eighteen members split into two equal groups of nine, eighty-one pairs cross between groups. There are 153 distinct pairs in total, so the score is eighty-one divided by 153, or about 0.529. The value is slightly above one half because a member is never paired with itself. With forty-eight members split equally, the corresponding maximum for two groups is about 0.511.

If every member receives the same label, separation is zero. If several balanced categories are occupied, it can be higher. A high score means that this particular sample of models produces diverse classified outcomes under the specified observation. It does not mean that the experiment has a large effect, nor that it is feasible or precise in an animal.

The denominator exposes a limitation. Suppose baseline recordings have already ruled out all silent models. A stimulus that separates silent models from active models may rank highly on the original grid but tell us little about the remaining biological possibilities. Experimental value should be assessed after conditioning on baseline-compatible members.

The ranking is also sensitive to duplication. If we add many nearly identical models in one region, the score changes even though we have introduced little new mechanistic diversity. Weighting mechanism classes or sampling from an explicit parameter distribution could reduce this problem, but each approach brings its own assumptions.

From separation to experimental value

A laboratory must care about more than the partition of a model grid. It must be possible to target the population, measure the predicted change, and distinguish the outcomes in the presence of biological variability and measurement noise. A model may predict two bump widths that are different numerically but indistinguishable with the available imaging method.

Interventions can also have effects outside the selected pathway. Silencing a cell type may alter several circuits simultaneously. The model's operation of setting a threshold extremely high is a clear computational manipulation, but its relationship to a biological manipulation has to be established rather than assumed.

A more mature ranking would predict distributions over measurable outcomes, incorporate intervention uncertainty, and evaluate information gain among models still compatible with prior observations. It could then account for cost and feasibility. The present score is a transparent first step toward that calculation.

Transparency is an advantage even when the score is simple. If a highly ranked experiment merely separates silent and active members, we can see that. If its ranking collapses when a classification threshold changes, we can see that too. An opaque confidence measure would make these weaknesses harder to diagnose.

Learning from an incomplete family

The ensemble method is most useful when it resists two temptations. One is to publish only the parameter setting that works. The other is to treat failure

throughout the sampled grid as proof that the anatomy cannot support the computation.

Between those extremes lies a productive result: a map of which assumptions produced which behaviours, together with the interventions that expose their differences. A model family can identify a narrow operating regime, reveal that baseline agreement hides causal differences, or show that an expected capability never appeared under the chosen protocol.

When the entire family fails similarly, the next step may be a new model class rather than a larger grid. The heading field and the APL discussion will make that possibility concrete. The ensemble has not exhausted the connectome's implications. It has organised a portion of them well enough that a disagreement can guide the next experiment.

Chapter 7

A Heading State in a Spiking Network

A heading representation has an unusual memory requirement. It must retain an angle while the animal is stationary and update that angle when the animal turns. It should also be able to use a visual cue without becoming merely a copy of that cue. These requirements make the heading circuit a useful place to distinguish persistent state, sensory filtering, and controlled movement of a population pattern.

The anatomical substrate highlighted here contains 148 cells across five named populations: forty-six EPG neurons, forty-two Delta7 neurons, eighteen PEG neurons, twenty PEN_a neurons, and twenty-two PEN_b neurons. The number eighty-eight refers only to EPG plus Delta7. The simulations themselves retain the full loaded network, as described in Chapter 6, while manipulating and reading these populations.

EPG activity is grouped by an eight-bin positional mapping. Delta7 supplies a strongly structured inhibitory path in the anatomical report. PEG participates in local recurrent copying, while the two PEN populations provide directionally offset feedback. This organisation motivates a ring-attractor hypothesis without fixing the gains or resting state needed to realise it.

What would count as a memory?

Imagine briefly illuminating a spot at one bearing. While the cue is present, a localised neural response could mean little more than that the sensory pathway has spatial tuning. A memory test begins after the localising input is removed.

The probe seeds activity near one position, releases the stimulus, and measures whether activity remains concentrated. It records the total response, circular concentration, and an estimate of width. It also drives the two PEN arms separately to measure directional effects. These observables are more informative together than a phase alone, because a phase can be decoded even from a weak or unreliable population pattern.

A completely silent population has no meaningful remembered direction. A uniformly active ring also has no preferred direction. A concentrated but almost inactive pattern can be difficult to distinguish from residual numerical or transient activity. The classification combines measures to separate these cases, though the exact boundaries remain choices of the probe.

Persistence over the observation window is a finite-time result. It is not a proof of asymptotic stability, nor a guarantee that the state will survive prolonged noise or arbitrary inputs. Those stronger claims require longer observations, additional initial conditions, and perturbations of the state itself.

Searching the operating regime

The forty-eight-member grid varies EPG recurrence by factors of one, three, four, and six. It varies the Delta7-to-EPG multiplier over one half, one, and 1.3. Tonic EPG bias takes values of zero, three, five, and seven millivolts. The PEN multiplier remains one.

At the recorded seed, thirty members are classified as silent, six as filters, and twelve as tonic-supported attractors. None is classified as an attractor without tonic bias. The successful settings occur with stronger recurrence and positive baseline excitation.

The twelve survivors represent one quarter of this grid. That fraction should not be read as an estimate of how often real flies have an attractor or how likely this mechanism is. The sample weights low and high gains according to the chosen grid, and the family holds many potentially relevant quantities fixed.

The result does establish a useful dependence in the tested model. Anatomically organised recurrence can support a persistent heading-like pattern, but the default operating point is insufficient under this protocol. Increasing excitability changes which computation the same network can express.

It does not establish that the animal must switch its compass on through tonic drive. A missing background input, a different membrane model, a more appropriate weight calibration, or an untested parameter combination could change the requirement. The experiment identifies tonic bias as an effective control in

this family. Physiology would have to establish what supplies the corresponding support in the fly.

Removing inhibition changes the explanation

The surviving models can look similar before perturbation while relying on different interactions. Delta7 silencing exposes this. Eight of the twelve retain a relatively sharp concentrated pattern under the classifier. Three lose the sustained response and are labelled as requiring Delta7 for the tested state. One loses confinement under the recorded outcome classification.

The last category needs particular care. The saved report calls it a confinement failure, while the current classification distinguishes that state from a surviving, widened bump. The original prose treated the one member as a broadened-bump success. That is not supported by the recorded class counts. No member in that report carries the separate widened-bump label.

This distinction matters because a later field model does produce a surviving broadened bump. It cannot be described as reproducing a one-member minority in the spiking report simply because both involve some loss of confinement. A broad coherent bump and broad or uniform activity with low concentration are different outcomes.

The three extinguished responses also need restrained interpretation. The circuit model contains active adaptation and depression, so disinhibition can interact with subsequent fatigue and recurrent feedback. But the label alone does not identify which process caused the shutdown. Removing adaptation while holding the rest fixed would test that explanation more directly. The present result is that the state was lost after the perturbation, not a demonstrated adaptation mechanism.

The value of the ensemble is visible despite these limitations. A baseline bump does not uniquely identify the interactions maintaining it. The perturbation produces several outcomes, and their continuous measurements can guide the next comparison. Majority vote is not a biological verdict; it reflects how the sampled settings are distributed among those outcomes.

Which perturbations split the grid?

On the recorded forty-eight-member grid, varying tonic excitation produces an unweighted separation score of 0.414. Lesioning the PEG copy pathway scores 0.370, and silencing Delta7 scores 0.351. Driving the right and left PEN arms gives lower and unequal scores, about 0.284 and 0.120.

The high tonic score is understandable. Baseline excitability is one of the dimensions that separates silence, stimulus following, and persistence in this family. A sweep across that dimension naturally divides the grid. It is useful for locating the operating regime, but its value after conditioning on a known persistent state could be lower.

PEG removal asks a more targeted causal question. Does the copy pathway provide necessary support, or can local EPG recurrence compensate? Delta7 silencing asks whether a particular inhibitory source is necessary for persistence or confinement. These questions can remain informative even when the animal's baseline activity has already excluded silent models.

The PEN asymmetry is also worth retaining. Oppositely oriented pathways need not have identical strengths or neuronal populations. A pooled directional score could hide a real difference between arms. Yet a one-sided model response can arise from the coordinate convention or stimulus as well as from anatomy, so both sign conventions and input conditions should accompany the result.

Earlier runs reported a recurring top-three set across seeds, with changes in order. That is evidence about those runs, rather than a general guarantee of ranking stability. A systematic sensitivity analysis would vary seeds, classification thresholds, and parameter sampling together, preferably after restricting to baseline-compatible models.

A second execution path

The same experiment was later expressed through the generic runner. At the same nominal seed it produced thirty silent members, eleven tonic-supported attractors, and seven filters. The best-ranked tonic experiment scored 0.393, and the leading set of experiments remained similar.

One member therefore crossed a class boundary. This is a useful compatibility result, but it is not exact reproduction. Nor does a small discrepancy prove independent confirmation. The implementations share major components, and random-number consumption, probe details, or classification can account for a difference. Establishing the cause would require matched streams and member-level comparison.

The scientific interpretation should survive this distinction. Both executions find a limited sampled regime of tonic-supported persistence and give prominence to excitation, PEG, and Delta7 manipulations. Claims about an exact fraction of viable models or an exact ordering of experiments are less stable.

What remains to be explained

The spiking network has recovered a family of heading-like responses attached to identifiable anatomy. That is a meaningful compatibility result. It has also revealed that the proposed computation depends on excitability and that different settings respond differently to inhibition loss.

The next questions are quantitative. How does width change as inhibition is gradually reduced? How does the velocity pathway move the state, and what gain relates drive to angular displacement? Which features of the response are consequences of the discrete wiring, and which belong to the chosen point-neuron dynamics?

A continuous field provides another way to ask these questions. It smooths away individual cells and exposes a mathematical description of bump shape and phase. That simplification will make some mechanisms easier to analyse, while creating new assumptions that must be stated just as explicitly as the assumptions of the spiking ensemble.

Chapter 8

The Shape and Motion of a Neural Field

A bump of population activity has more than one property. It has a position, a width, an amplitude, and a shape. When the animal turns, an ideal heading representation changes position while preserving enough of the other properties to remain readable. This suggests a second model in which the primary object is the activity pattern itself.

Instead of following each EPG spike, the field model represents activity as a function around a circle. Nearby positions can reinforce one another and distant positions can inhibit one another through spatial kernels. A stable bump is then a solution of a dynamical equation. Its movement can be studied as a perturbation of that solution.

This is an Amari-style neural field. It is an approximation with its own commitments: a regular spatial coordinate, repeated interaction rules, a chosen non-linearity, and a timescale. It does not inherit the authority of the cell-level reconstruction simply because its kernels were motivated by anatomical measurements.

Building the field

Let the field be called u , with angle and time as its arguments. In the implementation, the circle is sampled by 256 units. The activity relaxes toward rest, receives recurrent input, and responds to velocity and landmark signals. A saturating function converts the internal field into an output rate-like quantity.

The recurrence combines narrow excitation with inhibition. The narrow excitatory kernel represents local EPG and PEG support. The main inhibitory kernel follows the broad constant-minus-cosine profile motivated by the Delta7 pathway. A second, shallower inhibitory surround is included in the revised model.

In the implemented version, that additional surround is itself a constant-minus-cosine kernel with an adjustable depth. It is not merely a uniform subtraction of mean activity, although a uniform inhibitory component is one part of it. Its gain is independent of the Delta7 multiplier, allowing inhibition to remain when the Delta7 term is removed.

In mathematical notation, the equation can be written as

$$\tau \partial_t u = -u + W * f(u) + P + I.$$

Here W contains the excitatory and inhibitory kernels, the star denotes circular convolution, P is the selected velocity-related input, and I is the landmark input. The convolution sums contributions from other positions according to their offset. The nonlinearity f limits the output range, while τ sets the time over which the field responds.

Every term has a modelling rationale, but not every term is determined by the connectome. The anatomical profile motivates the shape of part of W . The local excitation's functional form, its gain, the saturation rule, and the field timescale remain assumptions. The second inhibitory term was introduced after consulting a perturbation result, as the next chapter explains.

A continuous family of positions

Suppose the unforced field has an equilibrium bump. If the kernels depend only on relative position, shifting that bump around the circle gives another equilibrium. The model has no preferred heading until a cue or asymmetry introduces one.

This translation symmetry produces a neutral direction. A small perturbation that simply shifts the bump does not restore it to the old position, because the shifted position is equally valid. Other perturbations, such as a change in width, may relax back toward the equilibrium shape.

The distinction is useful for heading estimation. The circuit needs a degree of freedom that can retain a new angle, while suppressing distortions that would make the angle ambiguous. An ideal attractor offers precisely this separation between phase and shape. Real finite circuits and numerical grids only approximate the symmetry, so position-dependent drift and pinning remain possible.

To derive a phase response, write the travelling profile as

$$u(\theta, t) = U(\theta - \phi(t)).$$

The symbol U denotes the equilibrated profile and ϕ its position. Differentiating with respect to time gives negative phase velocity multiplied by the spatial derivative of U . The minus sign follows from the argument being angle minus phase; reversing that convention would reverse the corresponding signs throughout the calculation.

The projection that phase reduction requires

Linearise the unforced field dynamics about U and call the resulting operator L . Translation symmetry implies that the derivative U' is a right neutral mode: applying L to it gives zero. But the vector needed to extract the phase response is generally the left, or adjoint, neutral mode.

Call that mode ψ . It satisfies the adjoint equation and must have a nonzero overlap with U' . To first order, a weak input of amplitude v and spatial profile P gives

$$-\tau \dot{\phi} \langle \psi, U' \rangle = v \langle \psi, P \rangle,$$

and therefore

$$\dot{\phi} = -\frac{v \langle \psi, P \rangle}{\tau \langle \psi, U' \rangle}.$$

The inner product sums or integrates the product of two profiles around the circle. This expression says that only the component of the perturbation that projects onto the phase-sensitive direction moves the bump to leading order. Other components can change its shape or relax away.

Using U' in place of ψ needs an additional argument. It is valid in an appropriate self-adjoint setting, but translation invariance alone does not make the linearised operator self-adjoint. Even a symmetric convolution kernel can be combined with a spatially varying derivative of the nonlinearity in a way that changes the adjoint relation.

The existing implementation uses the bump derivative in its projection and takes the magnitude of the resulting coefficient to set a positive gain. It does not solve the adjoint problem. Its reported gain near 0.27 should therefore be

treated as an approximate calibration within the specified field model, rather than an exact consequence of the wiring. The algebra in the earlier derivation also dropped the minus sign; the expression above retains it.

This correction changes the interpretation of the calculation without pretending that the implementation has already been replaced. A proper numerical check would compute the adjoint mode, compare the predicted response with small positive and negative inputs, and verify convergence as the grid and perturbation amplitude change. Those checks remain necessary before describing the phase gain as rigorously derived.

Why the equilibrium matters

Even an approximate projection should use the bump that the model actually sustains. A freshly imposed seed can be too narrow, too broad, or at the wrong amplitude. Its derivative then describes movement of a transient shape rather than the settled state.

The implementation allows a seed to relax before computing its gain estimate. Earlier calibration on the unrelaxed profile produced an inappropriate gain and a frozen tracking response. This is a useful example of a numerical procedure acquiring scientific significance: the object being measured was different from the object assumed by the derivation.

With literal shifted-feedback kernels and the approximate gain, the model tracked clean velocity at about 85 percent of the ideal rate in the recorded test. Shape distortion and finite-input effects are possible explanations for the deficit, but the unverified projection is another. It would be too strong to attribute all fifteen percent to higher-order effects before checking the adjoint calculation.

The gain also depends on the equilibrium, nonlinearity, timescale, and kernel normalisation. Changing any of them can change the numerical value while retaining the same underlying contact pattern. The phrase derived within a model is therefore essential even after the mathematical reduction has been justified.

Four mechanisms for moving the state

The project compared several ways to deliver velocity information. In the simplest shifted-input version, angular velocity changes the amplitude of a spatially displaced recurrent input. The intuition is that one side of the bump receives extra support, pulling the pattern around the ring.

This input does more than translate the pattern. It can also distort the shape. Under fluctuating velocity, repeated changes of sign can prevent those distortions from relaxing before the next input arrives. In the recorded single-channel benchmark, this mechanism had an RMS heading error of about 1.29 radians.

A second version introduced another field representing the PEN population while retaining amplitude-based drive. It did not repair the problem in the tested settings; the recorded error was about 1.61 radians. The presence of an additional dynamical population is therefore insufficient by itself. How that population represents velocity matters.

A third version used position. Velocity displaced a PEN-related bump, and the EPG-related field was pulled toward it. This reduced the recorded error to about 0.60 radians. The mechanism can express a phase difference between the two populations during rotation, providing a neural observable beyond final heading error.

The comparison supports position-based coupling within the tested implementations. It does not establish that every amplitude-based mechanism must fail, or that the animal uses precisely this two-field rule. Synaptic filtering was explored over a limited range, and the comparison shares the assumptions of the surrounding field model.

The ideal transport limit

The final version moves the field directly by advection. It shifts the activity pattern itself according to the velocity signal, approximating ideal transport. In the earlier single-inhibition benchmark, this version reached about 0.41 radians RMS error.

Advection is a useful engineering reference. It asks how well the field could track if phase transport were supplied directly. It is not evidence that the biological PEN pathway implements perfect transport. The distinction matters because the best later heading benchmark uses this mode.

The gap between position-coupled transport and advection is compatible with lag in the two-field mechanism, but it is a property of these model variants. Calling it the price paid by the fly would require establishing the biological coupling and matching its operating conditions. A simulated performance gap is not automatically an evolutionary tradeoff.

A stronger test would compare both behavioural and neural responses under matched inputs. Heading error would measure the task. The EPG-PEN phase relation, response lag, and bump deformation would test the mechanism. A model that tracks well but produces the wrong population relationship could

still be a useful estimator while remaining an inadequate explanation of the circuit.

What the field adds

The field makes it easy to measure a continuous width curve while changing inhibition. It also exposes the mathematical assumptions behind velocity updating. The spiking model retains irregular cell-level wiring and its particular adaptation rules; the field smooths these details into kernels and rate dynamics.

Agreement between them is useful when their assumptions differ in ways relevant to the question. Shared assumptions can also make both models wrong in the same way. Neither formalism deserves automatic priority. The next chapter examines a perturbation for which their disagreement helped identify an inadequate inhibitory description, while also showing why model revision and independent validation must remain separate.

Chapter 9

Disagreement, Revision, and a Fresh Test

A model becomes more interesting when it makes a prediction that could have gone another way. An ensemble makes several such predictions visible at once. But the scientific value of a comparison depends on when the evidence entered the process. A result used to revise a model cannot also serve as an independent test of the revised model.

The Delta7 investigation illustrates both the value of comparing formalisms and the importance of this boundary. The spiking ensemble and the field model gave different responses to reduced inhibition. Published physiology then motivated a change to the field. The revised field now makes a quantitative prediction that remains to be tested independently.

Keeping those events distinct produces a stronger account than compressing them into a story in which biology simply selected the winning model. The actual investigation included a failed simplification, a mechanistic revision, and a new uncertainty.

Before consulting the perturbation result

The spiking ensemble began with several baseline-compatible heading states. When Delta7 was silenced, most retained a relatively sharp concentrated pattern under the classifier. A smaller group lost the sustained response, and one lost confinement. As Chapter 7 explained, the recorded confinement-failure category is not the same as a surviving widened bump.

The original field attributed its surround inhibition entirely to Delta7. Under that construction, reducing Delta7 gain far enough caused the localised state to dissolve. In the reported exploration, the failure appeared below a gain of roughly 0.3.

This was a substantive model prediction. It arose because the field had one source of the inhibition needed to control recurrent excitation. Removing that source could eliminate the operating regime. The prediction was conditional on the field's architecture and parameter values, but it exposed a clear consequence of assigning all relevant inhibition to one pathway.

The models therefore disagreed even before deciding which outcome was biologically appropriate. Some spiking settings preserved a concentrated state; the single-channel field lost it under sufficiently strong suppression. That disagreement was useful because it directed attention to the representation of inhibition.

What the published observation establishes

Turner-Evans and colleagues reported that a single EPG bump can remain when Delta7 output is reduced, while its tracking of the animal's movements is impaired. They also argued that additional inhibitory sources must contribute to the circuit. This is the relevant qualitative observation for the comparison. [The neuroanatomical ultrastructure and function of a biological ring attractor](#)

The observation challenges a model that requires loss of the bump under a comparable reduction of Delta7 output. Comparability matters: a computational deletion of all output is not automatically equivalent to a partial experimental suppression. Matching the manipulation quantitatively would require knowing the residual effective output in the animal and representing it appropriately in the model.

The observation does not, on its own, decide whether a surviving bump should remain sharp or become wider. Survival and impaired tracking leave that distinction unresolved. Rejecting the sharp-survival spiking class would require a width measurement or an explicit test of tracking predictions under the perturbation.

This is a small logical point with large consequences. If a model predicts three features and an experiment measures one, agreement on that one feature does not validate the other two. Likewise, disagreement on an unmeasured feature cannot be used to reject a model. The observable must match the claim.

Adding an inhibitory pathway

The anatomical account already included inhibitory ring-neuron populations and their interactions with the compass. The field was revised to include a shallow inhibitory surround whose gain does not depend on the Delta7 multiplier. This provided a route for residual regulation after the structured Delta7 component was removed.

In the implementation, the second kernel has a gain of one half and a cosine-depth parameter of one half. Those values are model choices. Anatomy and physiology motivate the existence of additional inhibition; they do not uniquely specify this kernel or its numerical strength.

The revised model preserves a localised bump throughout the tested suppression sweep. At full Delta7 gain, its width at half maximum is about 90.5 degrees. Reducing the gain to 0.6 gives about 98.1 degrees; at 0.4, about 104.6 degrees; at 0.2, about 116.1 degrees. With the Delta7 term removed, the width is about 143.1 degrees.

Circular concentration falls from about 0.597 at baseline to about 0.528 at complete removal. Width and concentration are related but different observables. Width describes the span above half the peak, while concentration measures how strongly the population favours a direction. Both are needed to distinguish a broader coherent state from nearly uniform activity.

The revision therefore supplies a mechanistic explanation for survival within the field: a second inhibitory contribution continues to restrain the network. It also predicts graded broadening as the structured component weakens. The qualitative survival result informed the model's construction. The numerical broadening curve goes beyond that observation.

What is now a prediction?

The width curve can become a prospective test if it is fixed before comparison with an appropriate dataset. One would need graded perturbations, an estimate of effective suppression, and a consistent way to measure bump width. The model's full-width-at-half-maximum calculation should be related to the imaging observable, including sampling and indicator effects.

A model prediction expressed in degrees can look more precise than its measurement mapping warrants. Imaging reports a filtered signal from a finite set of regions, while the field provides an idealised activity profile. A fair comparison should apply a measurement model or at least quantify the effect of different width definitions.

Tracking offers another fresh test. The experiment reports impaired movement tracking after suppression. The revised model should therefore be driven through turns with and without the perturbation, measuring lag, drift, and phase errors as well as width. A bump can survive yet fail to update, so persistence alone is insufficient.

Several models with additional inhibition might reproduce survival. They could differ in the shape of the width curve, recovery after a transient perturbation, or response to simultaneous changes in excitation. Those differences are where the next discriminating experiment should focus.

The right status of the revised field is consequently clear: it is a model revised using a qualitative physiological constraint, with quantitative predictions awaiting a separate test. Describing that status does not diminish the revision. It prevents the same observation from being counted twice.

The cost of revision

The earlier single-channel field achieved an error near 0.41 radians in the recorded heading benchmark. The later two-channel model scored about 0.54 in the current benchmark configuration. Earlier drafts also reported an intermediate value near 0.449, reflecting a different recorded version.

These numbers show that the revised model was not simply selected to improve the headline engineering score. They do not isolate the causal cost of an additional inhibitory channel unless every other setting and input is held fixed in a matched comparison. Nor do they establish a precision-versus-robustness tradeoff in the animal.

A controlled ablation would compare the same benchmark trials under the same numerical settings, changing only the second inhibitory component and accounting for any necessary recalibration. It could then ask whether improved perturbation tolerance consistently reduces tracking precision within that model family. The biological interpretation would remain a further step.

This distinction is useful beyond the current example. When a model becomes more faithful to one observation and less accurate on another task, the result may reveal a real tradeoff, a poor parameter choice, or an interaction with the test regime. The discrepancy deserves investigation rather than an immediate evolutionary explanation.

Directional control and operating thresholds

Other field perturbations provide narrower compatibility checks. Under the tested directional input, retaining one PEN arm gives about 0.98 radians of drift over two seconds, while retaining the opposite arm gives essentially zero. This demonstrates arm selectivity in that implementation and stimulus direction.

The result should not be described as a complete bilateral validation. Reversing the input and testing both signs would be necessary to establish the corresponding symmetry, and biological pathway manipulations may have additional effects. A mechanism that is direction-selective by construction still needs to be checked against the response it is intended to explain.

An excitation sweep finds no bump at gains of 2.6 and below and a bump at 3.0 in the reported initialisation. This brackets a transition within the sampled values. It does not locate an exact bifurcation or establish that all initial conditions share the same threshold. Hysteresis would require approaching the regime from both directions and testing the dependence on initial state.

These qualifications point toward useful next experiments. They are ways to make a compatibility result more informative, rather than reasons to discard it.

Why the disagreement was productive

The comparison changed the question from whether a ring-like network can produce a bump to which inhibitory organisation can preserve and update it under perturbation. That is progress. The additional field representation exposed a simplification that was difficult to see from a baseline spiking trace alone.

The lesson is not that one formalism won or that an ensemble majority was defeated by a decisive width measurement. The recorded evidence supports a more specific account: the original field's inhibition was inadequate for the qualitative constraint, the revised field incorporated another pathway, and the resulting width and tracking predictions need independent evaluation.

A useful modelling system should preserve this history of evidence without turning the book into a changelog. The reader needs to know which observations shaped the model and which remain available to test it. Once that boundary is clear, disagreement between models becomes a source of experimentally tractable questions.

Chapter 10

When an Anatomical Offset Becomes a Computation

Suppose an input population is arranged in columns, and its connections consistently reach an output population several columns to one side. The most immediate interpretation is a shift: input at position six should produce output at position three. It is a simple and attractive way to perform a coordinate transformation.

But a map of contact counts is not an input-output function. The target neurons integrate several sources, interact with one another, and may send feedback to the input. A displaced projection can therefore provide a bias without determining the final response. The PFN-to-h Δ B experiment asks how much of the apparent shift survives those dynamics.

This is a useful second test of the operator approach because it asks a different question from the heading experiment. The observable is displacement along the selected column coordinate, rather than persistence around a ring. A framework that merely applies a heading classifier to every circuit would have little to say here.

Reading the structural offset

The structural histograms place the PFNd-to-h Δ B peak near negative three columns and the PFNv-to-h Δ B peak near positive two. With column six as the input, a literal reading would predict responses near columns three and eight respectively.

A histogram peak describes where contact weight is concentrated. It does not mean that every input neuron projects to exactly that offset, or that the output activity will have its maximum there. It also does not establish a rigid transform over all positions. Boundary effects and heterogeneous input can make a projection behave differently near the edge of the represented coordinate.

The two arms are anatomically unequal. The selected population contains forty PFNd cells and twenty PFNv cells. Their recurrent and feedback relationships also differ. A comparison based only on the location of two normalised peaks can conceal these differences in total drive and effective influence.

The initial dynamical probes often produced offsets with the expected sign but reduced magnitude, around one or two columns rather than the structural three. This is evidence that the static profile does not directly become the measured output displacement in those settings. It is not enough to identify why the displacement is reduced.

Possible contributors include the thresholded graph, other inputs, recurrence, saturation, and the centroid observable. The purpose of the deeper experiment is to separate some of these alternatives by changing the pathways that could support the response.

The loops behind the projection

PFNd has substantial self-recurrence, with about 7,970 synaptic contacts in the structural census. The graph also contains sixty-one $h\Delta B$ -to-PFN feedback edges, as well as $h\Delta B$ internal recurrence. These interactions make a simple feedforward interpretation incomplete as a description of the model.

The ensemble varies the gains of the two PFN-to- $h\Delta B$ pathways, $h\Delta B$ recurrence, and $h\Delta B$ tonic bias. Three gains for each PFN arm, four recurrent settings, and two bias settings produce seventy-two combinations. Each is tested with localised column drive and with selected recurrent or feedback pathways removed.

The recurrent ablations are causal tests within the simulator. They ask whether the observed shift depends on a pathway under the chosen operating point. They do not by themselves show whether the pathway carries the spatial transformation or merely keeps the response above threshold. Both can make a shift disappear when the pathway is removed.

This distinction can be tested by examining the full response. If an ablation abolishes all activity, loss of the centroid is different from a surviving response returning to the input column. A mechanism label that combines these cases

should be unpacked before making a strong claim about coordinate computation.

What was classified as shifted

At the recorded seed, thirteen members meet the shift criterion on at least one arm, fifty-five are classified as passthrough, and four are silent. The phrase shifted member therefore means that either PFNd or PFNv produced a response sufficiently close to its expected structural offset under the classification tolerance.

Only nine of the thirteen show the PFNd shift. The other four qualify through PFNv while their PFNd response is classified as passthrough. The hand-written mechanism analysis then examines dependency of the PFNd shift. Its nine entries account for those nine PFNd cases, not all thirteen shifted members.

Among the nine analysed cases, two depend on $h\Delta B$ recurrence together with $h\Delta B$ -to-PFN feedback, two depend on all three recurrent elements, and two depend on feedback alone. The remaining three each depend on a different single or paired combination involving PFN or $h\Delta B$ recurrence. None of these nine retains the classified PFNd shift after every tested source of recurrent support is removed individually.

That is narrower than saying none of thirteen shifted members can work from the projection alone. The four PFNv-only cases were outside this mechanism attribution. They must remain unassigned by that analysis rather than disappearing from the denominator.

It is also narrower than proving that the direct projection has no transforming role. The projection can supply the spatial bias while recurrence supplies sufficient activity or amplifies that bias. The experiment shows dependency on recurrent support in the analysed PFNd cases. Further probes would distinguish amplification from an independently generated displacement.

Is the shift the same everywhere?

A rigid translation adds the same displacement to every input position. If input at column four moves to column one, input at column six should move to column three, provided both positions lie within a valid coordinate range. A position-dependent displacement represents a different transformation.

The column sweep in the recorded experiment does not establish a uniform rigid shift across the driven positions. The measured output depends on which

column is stimulated. This is an important failure of the simplest operator interpretation under the chosen probe.

The observable itself deserves scrutiny. The hand-written probe computes a centroid over an ordered list of output columns and maps it back to a column label after rounding. If labels are not contiguous or the response is multimodal, a centroid in index space can differ from a centroid in the anatomical coordinate. Rounding can also move a response across a classification boundary.

A response concentrated equally at two separated columns illustrates the problem. Its centroid lies between them, possibly at a position where few cells are active. Calling that centroid a shifted copy can hide a qualitative change in the representation. The full profile, peak structure, and total activity should accompany the scalar offset.

For these reasons, the present sweep argues against claiming a clean rigid transform from the recorded model. It does not establish that the biological circuit cannot implement an appropriate coordinate transformation when driven by natural population patterns. Localised stimulation is a diagnostic probe, and the relevant input manifold may be broader than a single column.

The experiments that expose the differences

Varying stimulus amplitude has the highest unweighted separation score in the hand-written report, about 0.424. Removing $h\Delta B$ recurrence scores about 0.406, while removing $h\Delta B$ -to-PFN feedback scores about 0.316. Measuring the PFN_d offset and silencing PFN_v have similar scores, around 0.314 and 0.312. The rigidity sweep scores about 0.282.

The amplitude result is informative because a spatial operator must also have a usable response range. An output that appears shifted only for one input strength is a more conditional computation than a transform stable across a broad range. Saturation and recruitment can change both the magnitude and location of the response.

The feedback ablation is particularly useful for distinguishing a static reading from a recurrent one. It targets a path that the simplest feedforward interpretation would regard as unnecessary. If the shift or its supporting activity depends on that path, the proposed explanation must include the loop.

Again, the score is a property of the full sampled ensemble. Baseline-compatible shifted members are only a subset. Ranking interventions within that subset could change which experiment is most useful for deciding among the mechanisms that remain plausible after observing a shift in the animal.

What the generic execution adds

The generic runner reports eleven shifted members, fifty-six passthrough members, and five silent members at the same nominal seed. Amplitude scaling remains the highest-scoring experiment, with a separation of about 0.461. The counts are close to the hand-written results but are not identical.

The runner also includes a PFNV-silencing perturbation in its mechanism attribution. Some PFNd responses depend on PFNV co-activity under the tested conditions. This creates an arm-coupling hypothesis that the earlier dependency analysis did not address in the same way.

The perturbation's internal name can be misleading if read literally. A label recording that PFNV silencing changes the shift describes a dependency test; it does not mean the circuit needs the act of silencing PFNV to produce its normal response. Human-readable interpretation has to follow the direction of the intervention and the observed change.

Shared components limit how independently the two executions confirm one another. They agree on a broad picture of mostly passthrough responses and a small shifted subset, while exposing differences in classification and tested dependencies. A member-level comparison with matched input streams is the appropriate next step before attributing every discrepancy to harmless randomness.

A more useful interpretation

The anatomical offset remains a valuable constraint. It tells us where the direct pathway tends to send contact weight and motivates directional hypotheses. In the recorded model, however, the analysed PFNd shifts depend on recurrent support and do not establish a uniform input-independent translation.

The resulting hypothesis is that spatially biased projection and recurrent dynamics cooperate to shape the response. That statement can be tested more specifically by preserving total activity during an ablation, varying the width of the input pattern, and measuring the full output distribution across columns. These tests could distinguish recurrence that merely sustains firing from recurrence that determines the transform.

The circuit therefore illustrates a general advantage of executable hypotheses. A static offset gave a plausible first interpretation. Running the network exposed conditions that the interpretation omitted. The useful output is the revised set of dependencies and the remaining measurement questions, rather than a declaration that the wiring either does or does not contain vector arithmetic.

Chapter 11

A Negative Result in the Mushroom Body

The mushroom body is an especially tempting place to identify an algorithm from anatomy. Projection neurons expand into a much larger population of Kenyon cells. Kenyon cells receive inhibitory feedback and converge onto output neurons whose compartments also receive teaching-related input. The arrangement resembles a system for constructing sparse representations and learning how to read them.

A familiar architecture can make a failed simulation feel surprising. Surely, if the ingredients are all present, the expected operation should appear. But the expectation often bundles together several computations and several unmeasured physical assumptions. The mushroom-body experiment is useful because it forces us to take that bundle apart.

The implemented ensemble did not reproduce the chosen gain-control criterion. That is a result about a point-neuron family under direct Kenyon-cell stimulation. It does not identify synaptic conductance as the unique missing quantity, and it does not test the complete process of associative memory.

What the anatomical ratios mean

There are 4,064 Kenyon cells in the structural census. The broad projection-neuron count is 686, giving an expansion ratio near 5.9. Restricting the denominator to the 282 projection neurons that reach the calyx gives a ratio near 14.4. The familiar description of roughly sixfold expansion uses the broad denominator; it should not be attached to the calyx-projecting count.

The more important local fact is sparse sampling. A Kenyon cell receives input from a median of about five projection neurons. A large population of cells sampling different combinations can produce a representation in which similar sensory inputs activate partly different subsets.

That possibility depends on the transfer function. If every Kenyon cell responds linearly to any single active input, the expansion may mostly repeat correlations already present upstream. If cells require coincident input and operate near threshold, different combinations can become more distinct. Inhibition can help regulate that operating point, but its effect depends on how and where it is delivered.

The APL loop is anatomically extensive. About 196,000 APL-to-Kenyon contacts are distributed over roughly 4,210 connections, and reciprocal coverage reaches essentially all Kenyon cells in the selected graph. Dividing total contacts by cells gives about forty-eight contacts per Kenyon cell; dividing by connections gives a slightly different average. Neither quotient measures an effective inhibitory conductance.

The output population contains ninety-seven mushroom-body output neurons, each pooling many Kenyon cells. Teaching-associated dopamine pathways make compartment-specific contacts. These facts motivate a broad memory hypothesis, but a successful memory experiment would still need an experience-dependent change and a later recall test.

What the probe actually stimulates

The ensemble bypasses the sensory expansion by driving Kenyon cells directly. It creates two activity patterns, each involving two hundred cells with half of the selected cells shared. It then measures the similarity of the resulting output-neuron activity vectors and the response to stronger drive.

This is a reasonable way to examine the readout of overlapping Kenyon-cell patterns. It is not a test of whether projection-neuron input becomes sparse through Kenyon-cell expansion, because that transformation has already been supplied by the experimenter. Nor does the probe train the readout weights through experience.

The distinction is easy to see in a simpler example. If a camera-processing system contains an image encoder followed by a classifier, injecting a vector directly into the classifier tests the classifier's response to that vector. It says little about whether the image encoder would produce the same vector from a photograph. Bypassing a stage can isolate a question, but it cannot validate the bypassed computation.

The gain measure compares Kenyon-cell spike output at double drive with output at the original drive. Strong divisive regulation would tend to reduce that ratio below the unregulated increase over the relevant range. A low ratio alone would not prove normalisation, since saturation or fatigue can also compress responses. The effect of APL manipulation and the shape of the full input-output curve are therefore important.

The eighteen-member result

The ensemble varies Kenyon-to-output gain over one half, one, and two; APL gain over the same values; and output-cell tonic bias over zero and four millivolts. Output recurrence is held fixed. This produces eighteen parameter settings within the circuit builder's default LIF model.

Nine members are classified as silent and nine as collapsed under the selected response criteria. None satisfies the gain-controlled category. The separation score for both the baseline gain test and APL silencing is about 0.529, reflecting the same nine-versus-nine partition.

That score is not evidence of several successful competing mechanisms. It is the near-even two-class split explained in Chapter 6. If silencing APL leaves the classification unchanged, the classifier has not detected an APL-dependent change in the chosen capability. Continuous response differences could still exist below the categorical threshold.

A stronger diagnostic drives four hundred Kenyon cells at increasing input rates. At drive levels of sixty, one hundred and fifty, and three hundred, the recorded Kenyon-cell totals rise from about 4,163 to 9,144 to 14,338 spikes. The two model APL units produce roughly thirteen and eight spikes at the first level, then sixteen and eleven at the higher levels during the two-hundred-millisecond window.

Within this point-neuron configuration, APL output changes little while the driven population's output grows substantially. That helps explain why increasing drive is not effectively compressed through the model's feedback loop. It does not establish a biological limit of eighty hertz for APL, because the model has imposed a spiking representation on that cell.

Why the APL representation matters

APL is non-spiking, and its activity and inhibitory effects can be spatially localised. Amin and colleagues used local stimulation and calcium imaging to demonstrate that different parts of the cell can influence mushroom-body re-

gions differently. Broad anatomical coverage therefore does not imply a single global inhibitory signal. [Localized inhibition in the Drosophila mushroom body](#)

Representing each APL neuron as one spiking unit removes that spatial and graded organisation. Its output in the simulation is constrained by the point-neuron threshold, refractory behaviour, adaptation, and synaptic update rules. A plateau in the simulated spike count can diagnose those choices as much as it diagnoses the strength of the reconstructed contacts.

At least three explanations remain open for the failed gain-control test. The effective inhibitory weights may be inappropriate. The neuronal representation may omit relevant graded or local dynamics. The direct, strong stimulation may place Kenyon cells outside the operating range in which the feedback normally regulates them. These explanations can also interact.

The appropriate conclusion is therefore conditional: the tested point-neuron ensemble did not produce the specified gain regulation under this protocol. The experiment has located a mismatch between an expected operation and an implemented model. Identifying the cause requires comparing alternatives that differ in more than a single multiplier.

A comparison that could narrow the cause

A useful next study would keep the anatomical connectivity and input protocol fixed while changing the APL representation. A point-spiking version would provide continuity with the existing result. A graded version would remove spike-mediated output as a bottleneck. A compartmental version would allow local input to influence local output without forcing activity to mix uniformly across the cell.

The comparison should examine a range of stimulation strengths and patterns. Broad direct drive can be retained as a stress test, but more structured projection-neuron input is necessary to examine the sensory operating regime. Input width, overlap, and temporal structure can determine whether inhibition acts mainly near threshold or against strongly driven firing.

Conductance calibration should be varied within each representation rather than treated as an alternative to representation. A graded model with arbitrary weak feedback can fail, and a point model with very strong inhibition can pass a coarse compression criterion for the wrong reason. The goal is to find a model that explains multiple observables together.

Those observables should include Kenyon-cell recruitment, response rates, pattern overlap, the effect of APL suppression, and local versus distant inhibitory

influence where measurable. If only the total spike count is checked, a globally suppressive model can appear adequate while missing the structure of the code.

Learning and recall require another extension. The readout must be modified through a specified teaching rule, followed by a test on trained and untrained inputs. Until that is done, the ensemble remains a probe of fixed-pattern processing and regulation. The broad associative-memory operator is only partly examined.

Checking a negative result for a mundane cause

Before interpreting the APL failure biologically, the project checked the transmitter assignment. A diagnostic had displayed a missing value, raising the possibility that the inhibitory outputs had been omitted by the simulator's sign handling. Both APL cells were in fact labelled GABAergic in the relevant metadata path.

That check removes one possible implementation explanation. It does not validate the rest of the representation. A correctly signed inhibitory connection can still have the wrong magnitude, dynamics, or spatial scope. Debugging a negative result is a process of eliminating alternatives, not a ceremony after which the remaining preferred explanation becomes certain.

The same principle applies to stimulation and reset state. If a previous trial leaves adaptation or synaptic depletion behind, the next trial can appear suppressed for reasons unrelated to its parameter setting. Reproducible seeds help, but full state reset and clearly defined initialisation are equally important.

A useful negative result should therefore retain enough measurements to distinguish silence, saturation, and failure of the target operation. Reporting only that gain control was absent would discard much of what makes the experiment interpretable.

The apparent contradiction with whole-brain sparsening

A separate probe of the calibrated whole-brain model drives Kenyon cells through projection neurons. In that setting, suppressing APL raises the active Kenyon-cell fraction from about 0.15 to 0.72 and increases mean rate from about 0.87 to 5.08 hertz. APL plainly affects activity in that configuration.

The two findings concern different protocols and different model settings. The embodied calibration uses conductance-based synapses, altered thresholds and size scaling, and different adaptation choices. The circuit ensemble uses its own

defaults and direct Kenyon-cell input. The contrast cannot be attributed to stimulation alone while treating the models as otherwise identical.

It is nevertheless useful to distinguish sparsening from divisive regulation. A modest inhibitory effect can keep many near-threshold cells inactive without proportionally dividing the responses of cells driven far above threshold. A model can therefore affect the fraction of active cells while failing a strong-drive gain test.

To establish that this distinction explains the difference here, the same model would need to be tested under both input protocols with matched parameters. A crossed comparison of model configuration and stimulation route could separate the effects. The existing results motivate that study; they do not already supply it.

What the negative result contributes

The mushroom-body experiment is valuable because it exposes how much can be hidden inside an appealing computational label. Sparse expansion, inhibition, fixed readout, learning, and recall are related stages with different evidence requirements. A failure in one probe cannot stand in for a verdict on all of them.

The result also clarifies what additional information might matter. Synaptic efficacy remains relevant, but so do neuronal dynamics, spatial representation, and the operating regime. The next question is no longer simply how to increase APL gain until the category changes. It is which model of inhibition explains the relevant responses without relying on the wrong physical mechanism.

That is a more demanding outcome than a tidy positive result, and a more useful one than claiming that the connectome alone has identified its own missing parameter. The graph supplied the feedback anatomy. The experiment revealed the limits of one way of making that anatomy executable. The comparison among plausible alternatives is still open.

Chapter 12

What an Engineer Can Learn from the Heading Model

A biological explanation and an engineering algorithm can be evaluated separately. A field model may be an imperfect account of the PEN pathway while still offering an interesting way to combine noisy velocity with unreliable landmarks. Conversely, a model can reproduce a neural response without being the most efficient estimator for a machine.

The heading benchmark asks the engineering question. All estimators receive the same synthetic observations and attempt to recover the same trajectory. The result is preliminary evidence about their behaviour in one chosen regime, rather than a contest between the fly and classical estimation theory.

The most informative comparison turns out to be between a single stored angle and a distributed activity pattern. Both can represent heading, but they respond differently to conflicting observations. The benchmark explores whether that difference can be useful.

A deliberately difficult observation stream

Each trial lasts sixty seconds, sampled at intervals of fifty milliseconds. The true angular velocity fluctuates through a mean-reverting random process with occasional larger turns. The estimator receives a noisy velocity signal and a landmark bearing every two seconds.

Fifteen percent of landmarks are corrupted by an offset of exactly half a turn. This creates a conflict between the current estimate and an occasional observa-

tion pointing in the opposite direction. Every estimator sees the same trajectories and observations across sixty seeded trials.

The use of shared trials is a strength. A difficult trajectory is difficult for all estimators, so differences can be measured on matched conditions. It also determines the appropriate statistical analysis. The natural unit is the per-trial difference between estimators, not two unrelated collections of error scores.

The corruption model is special. An exact half-turn error lies at the antipode of a circular representation and may interact particularly strongly with a ring's inhibitory geometry. A convincing robustness claim should also survive a distribution of outlier angles, varying corruption rates, and genuine abrupt heading changes. Rejecting a bad observation is useful only if the estimator can still accept a surprising correct one.

The scalar estimators

Dead reckoning integrates the velocity signal and ignores landmarks. It has no way to correct accumulated drift, but it is also immune to landmark corruption. In this regime it provides a useful reference: a system that accepts unreliable corrections can perform worse than one that refuses them all.

A complementary filter also integrates velocity, then moves a fixed fraction of the way toward each landmark. The benchmark includes a correction fraction of 0.35 and a smaller fraction of 0.05. The latter is chosen to make the effective correction strength more comparable to the reduced ring estimator.

This matching matters because gain alone can explain an apparent advantage. A filter that reacts strongly to every landmark is vulnerable to corrupt observations. A cautious filter may look robust without doing anything more sophisticated than trusting them less. Comparing an attractor with only an aggressive baseline would confound architecture with parameter choice.

The Kalman-style estimator maintains an angle estimate and its uncertainty, using an assumed noise model to determine the correction gain. A gated version rejects landmarks whose discrepancy is too large relative to the estimated uncertainty. The gate provides explicit outlier handling, though its threshold and noise assumptions also affect performance.

The reduced ring stores a phase and applies bounded landmark correction. Once the distributed representation has been compressed this far, its relationship to a complementary filter is close. The comparison asks whether any meaningful advantage remains after that compression.

The distributed estimators

The full field retains 256 activity units around the circle. Recurrent excitation and inhibition maintain a preferred region, while landmark input competes with the existing state. The benchmark uses the revised two-inhibition field and idealised advection for velocity transport.

A second version adds independent noise to individual field units at the specified amplitude. This tests one form of population-noise sensitivity. It does not make the comparison with scalar estimators a fully matched sensor-noise experiment, because the additional perturbation acts on internal units of one representation.

The advection choice is particularly important. It supplies direct phase transport rather than requiring the shifted PEN-like mechanism to perform the update. The resulting score measures an idealised field estimator. Chapter 8's less direct transport mechanisms have their own errors and cannot inherit this benchmark result without being run under the same conditions.

The model also holds a landmark input for a finite interval. Cue duration and strength affect how much a competing activity peak can grow. Robustness therefore depends on temporal presentation as well as on the spatial kernel. Calling it purely structural would hide parameters that help determine the outcome.

Reading the recorded errors

The main score is RMS wrapped angular error in radians. Wrapping treats angles just below and above a full turn as nearby. The report also measures error in the five-second windows following corrupt landmarks, providing a focused view of recovery.

Dead reckoning has a recorded mean RMS error of 0.661 radians, with a standard deviation across trials of 0.285. Its post-outlier-window score is 0.589. The stronger complementary filter scores 0.724 overall and 1.176 after outliers, showing that these corrections can be worse than ignoring landmarks in this regime.

The ungated Kalman estimator has the largest recorded error, 1.159 overall and 2.044 after outliers. Adding a gate reduces the overall mean to 0.546 and the post-outlier mean to 0.784. This is a substantial change within the benchmark, and a reminder that the relevant classical comparison includes explicit handling of unreliable observations.

At the smaller matched correction gain, the complementary filter scores 0.622 overall and 0.630 after outliers. The scalar ring scores 0.653 and 0.650. Their trial standard deviations are about 0.211 and 0.228. The reduced biological description does not show an obvious engineering advantage in these aggregate numbers.

The full field scores 0.538 overall, with a trial standard deviation of 0.434, and 0.612 in the post-outlier windows. The noisy field scores 0.521 overall, with a standard deviation of 0.391, and 0.588 after outliers. These are the recorded means for the current benchmark configuration; the rewrite has not generated a new scientific benchmark run.

The field's overall mean is close to the gated estimator's 0.546. Its post-outlier mean is lower than the gated estimator's 0.784. Those observations motivate a recovery hypothesis. They do not establish statistical superiority from the summary alone.

Why paired uncertainty matters

Standard deviation across trials describes variation in trial difficulty and estimator performance. It is not a confidence interval on the difference between two methods. Large overlapping standard deviations neither prove a tie nor rule out a consistent paired advantage.

Suppose every difficult trial raises both methods' errors by the same amount, while one method is slightly better on almost every trial. Their separate standard deviations could be large even though the difference is stable. Conversely, a favourable mean could be driven by a few exceptional trials.

The next analysis should retain each estimator's score on each shared trial, compute paired differences, and report an interval obtained from those paired samples. The post-outlier windows also overlap when corrupt landmarks occur close together, so they should not be treated as independent observations. Trial-level resampling is a more natural starting point.

The slight improvement after adding field noise deserves the same restraint. It could reflect variation in a limited sample or a regime-specific effect. The recorded result shows no large deterioration at that noise setting. It does not establish that noise generally improves the estimator.

A possible mechanism for robustness

A scalar estimator can move directly toward an observation. In the field, a conflicting cue must alter a distributed pattern maintained by recurrent interac-

tions. A distant cue can begin growing a competing peak while the old peak continues to suppress it. A brief or weak contradiction may fail to displace the established state.

This provides a plausible explanation for reduced sensitivity to isolated outliers. It is an interpretation of the model's dynamics, and it can be tested by measuring peak competition directly. One could vary cue strength and duration while tracking whether the old state persists, a new state wins, or both coexist transiently.

The same persistence can become a liability. If the animal really turns quickly or a landmark legitimately changes the frame, resistance to correction can delay an accurate update. Robustness is therefore a tradeoff with responsiveness, not a free property obtained without tuning.

The comparison should examine this tradeoff across corruption distributions, velocity noise, landmark frequency, and cue duration. It should also retune competing estimators on separate training regimes rather than using the test results to choose favourable parameters. Otherwise the apparent architecture effect can partly reflect unequal optimisation effort.

The cost of keeping a population

The scalar methods store only a small number of values. The field stores 256 units and repeatedly applies spatial interactions. Even if its error is competitive, its computational cost differs substantially. Runtime and memory should accompany accuracy before recommending it as an engineering replacement.

The relevant cost depends on the deployment. A conventional processor may favour a compact gated filter. A parallel substrate or a system already maintaining a population representation may value the field differently. The present benchmark does not measure those costs, so it cannot settle the practical trade-off.

The useful conclusion is that spatial competition is a candidate mechanism for handling conflicting observations. The current field is competitive on this synthetic benchmark and has a lower recorded post-outlier mean than the gated comparison. Establishing the reliability, scope, and efficiency of that advantage requires paired statistics and broader regimes.

That result is sufficient to motivate further work. It allows the connectome-inspired investigation to contribute an engineering idea without claiming that a biological circuit has defeated an optimally designed estimator or that the best field transport mechanism has already been recovered from anatomy.

Chapter 13

Making the Investigation Reusable

Three successful analyses can still amount to three separate pieces of software. Each may contain its own population selectors, stimuli, classification rules, and result formatting. Adding a fourth circuit then requires reconstructing the same machinery again, with another opportunity for assumptions to drift.

The compiler claim becomes meaningful when circuit-specific reasoning is expressed separately from the common execution process. A detected operator should lead to a specification that states what is being varied and measured. The runner should then execute that specification without containing a new special-case experiment loop for every circuit.

This is an engineering boundary with scientific consequences. It makes repeated operations consistent and makes the circuit-specific choices easier to inspect. It does not automate the discovery of every relevant parameter or the interpretation of every result.

What belongs in a specification

Consider the mushroom-body probe. A specification identifies Kenyon cells as the stimulated population, mushroom-body output neurons as the readout, and APL as an inhibitory control population. It records the two-hundred-cell input patterns and their overlap. It identifies which connection classes receive variable gains and which output cells receive tonic bias.

It also explains why those quantities are free. Kenyon-to-output weights can change through learning in the biological system. APL efficacy is not measured

by contact count alone. Output recurrence exists, but its functional contribution is uncertain. Giving each axis a reason prevents the parameter list from becoming an arbitrary collection of convenient knobs.

The specification then describes interventions: suppress APL, remove output recurrence, or alter input strength. Finally, it names the observables and the rules used to group responses. The runner can enumerate the grid and execute the interventions, while the specification states the scientific question.

No source-code listing is needed to understand this separation. The essential idea is that the experiment is represented as data with a meaning. The same generic engine can interpret a different set of populations and gains without changing the logic of building, probing, perturbing, and reporting.

What the runner actually shares

The common execution machinery resolves populations from anatomical type labels, instantiates the chosen network, modifies selected edge classes, and applies tonic biases. It iterates over parameter combinations, resets state between probes, and records baseline and perturbed responses.

It also computes the unweighted separation score and assembles a common result structure. Every member remains associated with its parameters, observations, and labels. This matters because the summary should be recoverable from the underlying runs rather than being an independent narrative written after the fact.

The common builder retains the full loaded network. A specification selects the populations to manipulate and measure, rather than automatically extracting an isolated subgraph. That is a real design choice: background pathways can contribute to a response, and the computational cost remains that of the larger model. A future isolated-circuit mode would need an explicit boundary rule for omitted inputs and outputs.

The shared machinery also inherits a base neuronal configuration. As Chapter 6 established, it differs from the embodied calibration. Reuse is therefore not enough to guarantee that every stage uses the same physical assumptions. The effective parameter configuration needs to be recorded with each run, including defaults that are easy to overlook.

What cannot be shared without interpretation

The meaning of a response depends on the represented quantity. A heading probe needs circular statistics. A column-shift probe needs a coordinate map-

ping and a way to distinguish displacement from response loss. A memory-related probe needs pattern similarity, recruitment, and gain measurements.

These differences are handled by geometry-specific measurement components. The ring component estimates phase, concentration, width, and directional movement. The column component measures offsets and arm-specific responses. The pattern component compares activity vectors and drive-response relations.

A new operator can reuse the execution loop while still requiring a new measurement model. A motion correlator, for example, needs stimuli with controlled direction and speed, along with timing-sensitive responses. It would be misleading to declare support merely because the runner can stimulate its neurons and count spikes.

This boundary is scientifically healthy. It forces the question of what observation would establish the proposed computation. Automation should remove repeated bookkeeping, not remove the need to define the observable correctly.

The column-centroid issue from Chapter 10 illustrates the risk. A generic function can be perfectly reusable while computing the wrong coordinate average for a particular anatomy. The geometry component needs its own checks, including examples whose expected result can be reasoned through independently of the implementation.

Comparing the two routes

The hand-written heading lab produced thirty silent, twelve tonic-supported attractor, and six filter members. The generic execution produced thirty, eleven, and seven. Both gave prominence to tonic excitation and the PEG and Delta7 interventions, while their precise scores differed.

The PFN lab produced thirteen shifted, fifty-five passthrough, and four silent members. The runner produced eleven, fifty-six, and five, and included an additional arm-coupling dependency in its attribution. The mushroom-body experiment was first executed through the generic pattern component and produced nine silent and nine collapsed members.

These comparisons support approximate consistency of the broad outcomes. They do not establish numerical equivalence. A difference of one or two members can be scientifically minor if it reflects a threshold boundary, but that explanation should be demonstrated from the member-level observations rather than assumed.

Shared code also limits independence. Agreement can show that the specification expresses the intended experiment, but it cannot independently verify the neuron kernel both routes use. A deterministic reference calculation or a separate implementation of a small case would test a different part of the system.

Exact agreement would not be suspicious by itself. It could be the expected result of identical inputs and operations. Likewise, small disagreement does not provide evidence of desirable independence. The cause of agreement or disagreement depends on what is shared and what differs.

Preserving the argument with the result

A useful computational record contains more than the final winning experiment. It includes the seed, parameter grid, effective neuronal settings, selected population counts, coordinate convention, stimulus timing, continuous observables, classification thresholds, and perturbation outcomes.

The present artifacts retain many of these elements, including parameter combinations, classifications, ranked experiments, and selected cross-formalism comparisons. They do not yet eliminate every need to inspect the source for effective defaults or measurement details. The discrepancy between the prose and the ensemble builder shows why those omissions matter.

A generated report can reduce such drift. Population counts and class totals should be read from the same run being discussed. Mechanism summaries should state which subset they cover. If an analysis attributes mechanisms only for one arm, the denominator should remain visible when a broader hypothesis count is displayed elsewhere.

This is a place where software discipline directly improves scientific communication. A repeated number should have one computational origin. A report can then translate it into prose without manually maintaining conflicting copies. The reader should encounter an explanation, while the supporting record carries the exact accounting.

How much has been compiled?

The system now has a real lowering boundary from candidate operator to an executable specification and then to an ensemble report. Common execution supports several distinct geometries. Adding an experiment within an existing geometry can largely become a matter of specifying populations, gains, and perturbations.

The choices of free axes and plausible ranges are still human-authored. So are the operator definitions and much of the interpretation of their outcomes. Automatically turning every edge class into a parameter would be easy to express and often impractical to execute. It would also assign complexity according to annotation granularity rather than biological uncertainty.

A better automatic proposal system would group parameters using physiological evidence, identify sensitivities, and refine the sample where predictions change. It would distinguish uncertainty in a measured quantity from uncertainty about the model class. These are substantive modelling problems, not the final few lines of routine automation.

The current compiler is therefore a reusable system for executing specified connectome-constrained hypotheses. It is not an autonomous discovery engine that recovers the complete programme of a nervous system. Its value is that a hypothesis can move through a common, inspectable process and return with its assumptions and failures attached.

The next boundary to improve

Before adding many new operators, it would be useful to strengthen the contract of the existing ones. The same input should produce a traceable set of effective parameters. The same intervention should have a clearly defined mapping to an observable. Numerical differences between execution routes should be explained at the level where they arise.

That work may be less visible than detecting a fourth circuit, but it increases the value of every future result. A system that can execute many hypotheses quickly is useful only if the claims remain connected to what was actually executed.

The embodied model raises the same issue on a larger scale. There, several neural, sensory, physical, and behavioural components cooperate to produce movement. The next chapter follows those boundaries through the whole system, treating the moving fly as a demanding integration experiment rather than a substitute for circuit-level explanation.

Chapter 14

Closing the Loop Through a Body

A stationary neural model receives inputs chosen by its experimenter. An embodied model helps choose its own next input. A motor signal moves a joint, the foot touches the ground, and a sensory pathway reports the contact back to the nervous system. The next motor signal now depends partly on the consequences of the previous one.

This closed loop can reveal failures that an isolated stimulus never exposes. A response that is useful once can become unstable when it repeatedly stimulates itself. A visual pathway that detects approach can also respond to the animal's own moving limbs. A small persistent steering bias can become a large circular trajectory.

The embodied fly is therefore a demanding integration test. It combines the reconstructed graph with a neuron model, a sensory front end, a physical body, motor readouts, and several supplied behavioural mechanisms. Understanding the demonstration means following how those components contribute, rather than assigning the whole animation to the connectome.

The assembled animal

The neural system represents all 165,122 neurons in the packed dataset. The embodied fit uses conductance-based integrate-and-fire dynamics with a half-millisecond timestep and a connection threshold of six contacts. The larger three-contact graph remains the data source, but not every stored connection contributes under that runtime threshold.

The physical body comes from flybody, a MuJoCo model with sixty-seven bodies, 102 joints, and seventy-eight actuators. Its mass is about 0.98 milligrams, and the physical simulation uses a 0.2-millisecond step. The model includes

adhesive tarsal contact, which is important for a small animal whose feet must maintain purchase on surfaces.

Vision comes through a trained flyvis optic-lobe model with a 721-column hexagonal lattice per eye. Its outputs are mapped onto approximately 62,000 corresponding neurons in the male-CNS representation by cell type and retinotopic position. The training embedded in this visual front end is part of the assembled model's prior information. Visual competence cannot all be attributed to the untrained central graph.

Other sensory inputs include taste, smell, contact, proprioception, wind, and heat. They enter as modelled spike trains on 7,745 sensory neurons across 151 channels, using rates motivated by physiological sources. Motor readouts use identified descending populations and a mapping of 439 motor neurons to 170 muscle groups.

Those mappings are scientific assumptions as well as engineering interfaces. A rate-to-force conversion can alter posture. A sensory-rate conversion can alter the likelihood of an escape. Their calibration should be understood alongside the neural weights, because all participate in the causal loop.

Making the graph executable

The embodied fit adds several properties not specified by contact counts. Synapses use excitatory and inhibitory reversal potentials, with the inhibitory value fitted near negative 76 millivolts. A postsynaptic size correction scales the effect according to volume relative to a regional reference, with an exponent near 0.61.

Kenyon-cell thresholds are raised by about 10.9 millivolts relative to the base setting. Lamina cells receive a resting bias near 5.3 millivolts so that inhibitory photoreceptor input can modulate their activity in the intended regime. Sensory cells are prevented from being directly driven to spike by central feedback in this simplified receptor model.

A known electrical connection associated with the giant-fibre jump pathway is supplied separately because it is absent from the packed chemical graph. Slow octopamine modulation is also implemented explicitly. These additions are motivated, but they remain additions to the information contained in the graph representation.

The fitted configuration uses a synaptic scale near 0.55, inhibitory gain near 0.61, and refractory period near 3.8 milliseconds. Spike-frequency adaptation is zero in this fit. These values differ from the default circuit ensembles, which

is why behaviour across the two branches cannot be compared as though only the body had changed.

Several explored alternatives failed useful checks. A whole-network rate model produced large waves of activity without a coordinated stepping rhythm. Adaptation and short-term depression could stabilise some activity while weakening the sugar-to-proboscis pathway. Treating uncertain transmitter assignments as uniformly excitatory caused excessive olfactory spread. These are findings about the tested implementations, not universal arguments against rate models, adaptation, or excitatory uncertainty.

Calibration and its boundaries

Nine global parameters were searched against a suite of physiological and behavioural targets. The recorded search used twenty generations of twenty-four candidates. Its objective combined several response measures rather than fitting a single attractive demonstration.

The selected model produces about fifty-nine hertz in the proboscis motor neuron under labellar sugar stimulation, while bitter stimulation and a sugar-bitter mixture suppress that response in the measured protocol. Front-leg sugar produces about thirty-four hertz in the same motor readout. Leg sugar reduces the forward-walking drive from about 9.2 to 4.6 hertz and produces little backward-walking activity.

Kenyon-cell recruitment reaches a 7.4 percent target in one calibration probe. Takeoff-related neurons respond more strongly to the modelled loom than to self-motion, roughly thirty-three versus two hertz in the recorded fit. Direct activation of a walking-associated descending pathway reaches leg muscle groups.

These are calibration outcomes. The observations used to select parameters should not also be described as independent validation of those parameters. They establish that the chosen model can satisfy parts of the objective under the fitting protocol. Generalisation requires other stimuli, states, and perturbations.

The best recorded composite score is 0.746, with a mean around 0.725 over six evaluations. An earlier configuration had a higher isolated score but a lower repeated mean. A composite objective is useful for optimisation, but its numerical scale is not a percentage of biological correctness. Its interpretation depends on the weighting and form of its terms.

Some failures remain visible even in the selected fit. Odour specificity in projection neurons and Kenyon cells failed across the searched configurations. The saved calibration also records background activity and a nonzero baseline

proboscis-motor response, so the successful evoked response should not be read as complete resting-state fidelity.

The olfactory failure may involve broad cholinergic local-neuron pathways, but the search does not isolate that explanation. Missing receptor-specific effects, inappropriate dynamics, or sensory stimulation could contribute. What it establishes is that the tested global parameterisation did not preserve the desired specificity.

When an objective rewards the wrong solution

Two implementation failures encountered during calibration are worth understanding because neither requires exotic biology. An odour-specificity objective could be satisfied by making Kenyon cells silent. If no cells respond, two odours do not share an active pattern, but the system has not produced a useful sensory representation.

A specificity measure therefore needs a response constraint. It should reward distinct informative activity, rather than merely low overlap. This is a general problem whenever a desired property has a trivial degenerate solution. A sparse code can be too sparse to carry information; a stable network can be stable because it never responds.

A separate issue involved reused simulation memory retaining activity counts between evaluations. Without a complete reset, one parameter candidate could inherit part of another candidate's state. The resulting score would appear stochastic or state-dependent even though it reflected an evaluation bug.

These checks are part of the experimental method. An optimisation process is good at finding weaknesses in its objective and implementation. A plausible final score becomes more credible when the obvious degenerate routes have been examined, but passing those checks does not replace independent biological tests.

Reafference changes the operating point

With the body held still, the forward-walking descending populations were largely silent in the relevant probe. Allowing the body to move produced foot-contact bursts that drove them to around ten to thirteen hertz. The simulated fly then ran persistently. Masking tarsal contact reduced the same drive to roughly three hertz.

The network was responding to the consequences of movement. This is reafference: sensory input generated by the animal's own action. It can be useful

feedback, but in this model it also reinforced locomotor drive strongly enough to dominate the behaviour.

The implementation therefore attenuates footfall bursts during stepping, using a suppression factor tied to step amplitude. This resembles a plausible form of sensory gating, but the numerical attenuation is supplied by the model. The reconstructed contacts did not specify its gain or state dependence.

Constant food contact created a related problem. Persistent tactile drive could keep the animal moving rather than allowing it to stop. Introducing adaptation in the tactile input made sustained contact less effective than a new contact. Again, the correction changes the sensory dynamics at the interface, and its contribution should remain visible when interpreting feeding behaviour.

The steering readout exposed asymmetry. Right-side descending populations received different excitation and inhibition from their left-side counterparts, and the uncorrected model tended to circle. A slow adaptation with a four-second timescale was added to the steering readout. The observed circling belongs to the interaction between asymmetric wiring and the chosen dynamics; it does not uniquely diagnose an anatomical defect or a biological turning preference.

Commands are not motor programmes

The network can propagate stimuli to identified descending populations associated with forward walking, backward walking, steering, grooming, escape, take-off, and courtship. This is useful evidence about pathway recruitment. It is not the same as producing the complete corresponding movement through raw motor-neuron output.

Walking makes the distinction unavoidable. In the tested full-connectome motor mode, directly using nerve-cord output to drive leg muscles did not maintain posture or coordinated locomotion. That failure applies to this model and motor interface. It does not establish that rhythm or posture are absent from the animal's wiring.

The working walking mode uses a stepping generator fitted to one hundred recorded trajectories. Its source movements have a median step frequency around 9.5 hertz and speed around 1.7 centimetres per second, with the fitted joint trajectories reporting roughly twelve degrees RMS error. Descending signals influence the generator, which supplies the detailed coordination needed by the legs.

A video of the resulting fly walking therefore demonstrates an integrated neural-command and movement-generation system. To claim that the connectome

generated the rhythm would require replacing or disabling the supplied generator and showing the rhythm in the neural motor output under an appropriate body model.

The same distinction applies to escape and righting. Explicit movement programmes support the jump and wing-assisted recovery. The jump landed upright in twenty-four recorded trials, while wing-assisted righting succeeded in four of six inverted starts. These are small, configuration-specific tests of supplied motor programmes, rather than evidence that those programmes were inferred from the connectome.

Where spontaneous activity comes from

The model includes an endogenous process that schedules bouts, pauses, grooming, saccades, and local search. Its statistics are motivated by behavioural literature: walk bouts around a 2.2-second median, pauses around 1.4 seconds, and saccades occurring during walking with a tendency to alternate direction.

This process delivers drive to identified descending populations rather than writing actuator commands directly. That keeps the neural network in the signal path, but it does not make the timing of bouts connectome-derived. The network mediates a supplied schedule.

This distinction is worth dwelling on because it applies to many demonstrations of embodied intelligence. A system can pass every motor command through a neural model while an external process still determines when to walk, when to stop, and which action to favour. The origin of the command matters as much as the route through which it passes.

In the embodied operating state, descending neurons receive substantial inhibitory conductance. A nominal current-like bias can therefore have less effect than expected from the resting model. The project uses excitatory conductance to deliver the endogenous drive more effectively. This is a reasonable interface adjustment, and it is another reminder that the operating point changes when the loop is closed.

The failure to obtain spontaneous bouts from the unaided model does not show that bout structure is absent from anatomy. It shows that the selected neuronal dynamics, inputs, and state variables did not generate the desired statistics. The missing explanation could involve a parameter, an omitted state process, or a more appropriate circuit model.

Escape in a moving world

Looming detection is harder once the animal moves. A wall approached by the fly expands on the retina, but so can a grooming leg. A pathway tuned to image expansion may therefore fire during self-generated movement unless other information changes its interpretation.

The implementation adds gating based on contact, optic flow, and action-related signals. Without sufficient gating, the fly can repeatedly trigger escape near walls or during grooming. With strong gating, genuine external looms can be missed. The tradeoff is exposed by closing the sensory loop.

In one recorded test, only two of ten looms produced escape. Takeoff-neuron activity during self-motion and grooming could approach the trigger level, while real looms provided only a limited additional margin. This is a weak discriminator in the tested system, even if the isolated calibration shows a larger loom response than a particular self-motion stimulus.

The next useful work is pathway-specific: examine the visual inputs, giant fibre, and takeoff populations under matched naturalistic and self-generated stimuli. A single global score cannot identify which stage loses the distinction. The existing model helps locate a problematic chain without having established its unique weak link.

A state signal that does not explain its phenotype

Octopamine provides a particularly instructive example of an anatomically motivated addition. The model links nutritional state to hormonal and neural signals, then uses octopamine-related activity to lower target thresholds over a slow timescale. About 12,851 anatomical targets receive this simplified modulatory effect.

The model does not have a receptor map specifying which target should change in which way. It applies a common effect, capped near two millivolts, and calibrates the resting activity of the modulatory populations. Fast octopamine contributions are removed as part of this modelling choice. That choice should not be generalised into a claim that every octopaminergic neuron lacks every fast or co-transmitted effect.

With the full state-aware behavioural system, the recorded wild-type moving fraction rises from about 0.55 when fed to 0.73 when starved. The no-octopamine-synthesis condition rises only from about 0.51 to 0.56, while the AKH-receptor condition changes little, from about 0.64 to 0.65. These qualita-

tive patterns resemble the targeted phenotypes, but the bout process can read the arousal state and contributes to the match.

When that behavioural process is made blind to arousal, direct modulation of the graph does not reproduce the intended starvation-induced increase. The recorded moving fraction instead changes from about 0.60 to 0.57, with a lower walking speed in the starved condition.

This is an important negative. The uniform threshold-modulation model is insufficient to produce the target phenotype through the network alone in this setup. The anatomical distribution of contacts may contribute, since steering and backward pathways receive substantial modulation, but the experiment does not isolate that distribution as the cause. Receptor-specific effects, dose, dynamics, and the motor readout remain alternatives.

Removing fast octopamine contributions also weakened the loom pathway in the recorded configuration. Subsequent fitting recovered some takeoff response without recovering the giant-fibre response. An added locomotion-associated visual modulation did not resolve the problem either. These interventions rule out particular attempted repairs; they do not uniquely localise the biological mechanism that is missing.

What the moving model establishes

The project has shown that a large reconstructed graph can be connected to a physical body and sensory environment, and that this system can express several plausible pathway responses and supplied movements. It has also exposed failures of specificity, coordination, self-motion discrimination, and state dependence that are much harder to see in a static diagram.

Its runtime is an engineering result with a hardware-dependent scope. Earlier measurements reported roughly one fifth of real time per fly with vision and several simultaneous flies on a twelve-core laptop. These figures describe that implementation and machine, not a timeless performance property.

Cross-backend checks provide partial implementation evidence. A deterministic chain gave matching spike counts in the two tested kernels, while a longer stochastic drive produced differing totals, about 10,298 versus 9,580 sampled spikes. One close aggregate comparison is not proof of stochastic equivalence; matched streams and distributions over repeated trials would provide stronger evidence.

The scientific value lies in the boundaries exposed by the integration. We can distinguish pathway recruitment from movement generation, state-aware bout

scheduling from direct modulation, and isolated sensory responses from closed-loop behaviour. Each distinction suggests a more targeted experiment. The body makes the model answer harder questions, while the circuit analyses help explain which part of the answer came from which assumption.

Chapter 15

What the Wiring Has Taught Us to Ask

A complete wiring diagram changes the questions available to neuroscience. It lets us replace some conjectures about connectivity with measured structure. It identifies pathways that can be manipulated, reveals repeated organisation, and supplies a common anatomical substrate for competing dynamical explanations.

It does not select a unique executable programme. The same graph can support different behaviours when its synaptic efficacy, cellular dynamics, background inputs, and state change. Some of those possibilities can be sampled in a model ensemble. Others require a different representation before they can be expressed at all.

The investigation in this book has used the connectome to organise that uncertainty. Its most defensible achievement is a chain from anatomical pattern to candidate computation, from candidate computation to explicit model family, and from model family to perturbations that expose differences. The chain is incomplete, but its boundaries can be stated.

The recurring gap

Several results begin with a compelling structural signature and end with a more conditional dynamical claim. The Delta7 path has an excellent cosine fit in the structural measure, while a particular realised inhibitory probe has a weaker fit. Some heading-model settings maintain a localised state, but none in the reported grid does so without tonic support. The PFN projection has sys-

tematic offsets, while its analysed PFNd shifts depend on recurrent support in the simulation.

The mushroom body has extensive feedback anatomy, yet the point-neuron ensemble fails its chosen gain-control test. The embodied system recruits plausible command pathways but needs supplied stepping and bout mechanisms to produce its demonstrated movement. These are related observations, though they do not all diagnose the same cause.

The phrase structure systematically overstates what dynamics delivers can serve as a warning about our expectations. It should not become a claim that the biological graph promised more than the biological dynamics can accomplish. The graph itself makes no verbal promise. We interpreted a pattern as an operation, then tested that interpretation using a model.

When the operation failed to appear, at least two things were under examination: the proposed computation and the way the anatomy had been made executable. A false interpretation, an unmeasured gain, a missing mechanism, and an unsuitable probe can all produce a negative result. The useful next step is to design a comparison that distinguishes them.

What has been demonstrated

The structural analyses can operate on a common representation of fly and worm connectivity. They expose recurrence, sign-conditioned motifs, and concentrated spectral modes. Their cross-species comparison remains dependent on differences in annotation, thresholding, and sign coverage, but the common interface makes those differences inspectable.

The ensemble machinery can execute parameter grids for several candidate computations and retain the outcomes of defined perturbations. It demonstrates that baseline-compatible responses can rely on different interactions. The unweighted separation score makes these disagreements easy to summarise, while leaving experimental value and sampling sensitivity as further problems.

The field model supplies another description of the heading system. It produces continuous width and phase observables and exposes the assumptions required for a phase-reduction argument. Its inhibition was revised using physiological evidence, and its resulting quantitative predictions remain available for a new test.

The compiler has a reusable specification and execution boundary. Circuit-specific choices can be separated from common iteration and reporting. The remaining human choices are substantial, including the operator vocabulary,

free parameters, model class, and measurement geometry. Reuse has made the investigation easier to extend; it has not made those choices automatic.

The engineering benchmark shows that an idealised field estimator can be competitive in one synthetic regime. The embodied model shows that connecting the graph to a body creates informative failures and useful pathway-level responses. Neither result substitutes for a validated explanation of the whole animal.

Missing quantities and missing descriptions

Some information is plainly absent from the packed graph. Contact counts do not provide receptor-specific conductances. The chemical representation does not include a complete electrical graph. It does not supply each cell's membrane properties, all synaptic timescales, or the distribution of modulatory receptors. Thresholding also removes weak connections that may matter collectively.

Other limitations arise from the chosen representation. A neuron stored as one node can have spatially separated computations. A type-level edge can hide heterogeneity among cells. A field kernel can erase irregularities that stabilise or destabilise a real population. A centroid can collapse a multimodal response into a position at which little activity exists.

These categories require different remedies. Measuring conductance can narrow a parameter. Modelling local graded activity can change the class of mechanisms. Improving a coordinate map can correct an observable without changing the simulated biology. It is important to identify which kind of uncertainty a proposed experiment addresses.

APL is the clearest example. The failed direct-drive gain test does not isolate a single missing scalar. It motivates a comparison of effective strength, neuronal representation, and input regime. The result is scientifically useful precisely because it reveals several distinguishable explanations rather than prematurely choosing one.

The status of a prediction

A prediction is strongest when its assumptions and observable are fixed before the relevant data are examined. This need not mean that a model is built without any prior biological knowledge. It means that the observation used as a test was not also used to select the tested answer.

The Delta7 revision illustrates the distinction. Qualitative survival after suppression motivated an additional inhibitory contribution. Agreement with survival

is then a consistency check on the revision. The detailed width curve and the model's altered tracking response can provide fresh tests if specified before comparison with suitable measurements.

A similar boundary applies to calibration. The embodied model was fitted against several reflexes and activity targets. Reproducing them shows successful fitting within the objective, while new perturbations and closed-loop conditions test generalisation. The benchmark's parameter choices likewise need a separation between tuning regimes and evaluation regimes.

Maintaining these boundaries is not merely cautious language. It changes which result should be pursued next. Once an observation has shaped the model, repeating that agreement yields less information than measuring a consequence that the revision introduced.

How to make the current account harder to dismiss

The next work should deepen the existing comparisons before multiplying operator names. For the heading field, the phase calculation should use the appropriate adjoint mode and be checked against small signed perturbations. This would turn an approximate within-model calibration into a justified reduction, or reveal the limits of that reduction.

For the mushroom body, comparing spiking, graded, and compartmental inhibitory representations under matched inputs would test the competing explanations of the gain failure. A complete sensory expansion and a learning-and-recall protocol would then examine the broader memory hypothesis that the current probe leaves untouched.

For Delta7, a fresh quantitative perturbation dataset could test width and tracking together. The suppression level should be related to effective synaptic output, and the model should be observed through a measurement procedure comparable to the experiment. A surviving bump with the wrong update dynamics would be a partial success, not a complete one.

For the PFN circuit, the four PFNv-only cases deserve their own mechanism attribution. The coordinate readout should operate directly on the stated anatomical coordinates, preserve full response profiles, and distinguish loss of activity from loss of displacement. Matched-amplitude ablations could then clarify whether recurrence computes the shift or supplies the activity needed to express it.

The experiment-ranking method needs a corresponding refinement. Baseline-incompatible models should be excluded or downweighted, and rankings

should be tested against resampling and class-boundary changes. Measurement noise and feasible intervention strength should enter before the score is presented as laboratory priority.

The engineering benchmark can be strengthened with paired uncertainty intervals, distributed outliers, true abrupt changes, separate tuning trials, and run-time and memory measurements. These additions would say whether the apparent recovery advantage is reliable and whether it is worth its computational cost.

A different way to value a connectome

One way to judge a connectome is to ask whether it lets us simulate an animal from scratch. That is an ambitious and worthwhile goal, but it compresses many sources of missing information into a single dramatic test. A failed walking simulation then seems to say that the wiring was insufficient, even when the failure might be in the motor interface or cellular dynamics.

Another way is to ask how the connectome changes the space of explanations. It can rule out a proposed direct pathway, identify an unexpected feedback route, or show that two candidate mechanisms act on the same anatomical substrate. It can turn a vague question about inhibition into a perturbation of named populations.

This value appears before a complete simulation succeeds. It also survives a model's failure. The PFN investigation gained an explicit recurrence question; the APL investigation gained a representation question; the heading investigation gained a quantitative test of residual inhibition. Each is a more constrained scientific problem than the one with which it began.

The long-term ambition is to make this process systematic. A connectome would enter with its uncertainties, generate a set of plausible computations, and produce experiments that reduce the uncertainty efficiently. Evidence would then revise both the parameter distribution and the model classes considered.

The current project has built part of that loop. Its next advance will come from making a few of its predictions survive more demanding comparisons. A nervous system is not understood when a diagram receives an algorithmic name. Understanding grows when the name leads to a mechanism, the mechanism leads to a measurable consequence, and the consequence survives a test that could have shown it wrong.

Appendix A

Reproducing the Investigation

A numerical claim is most useful when a reader can recover the experiment that produced it. The prose explains what a result means; the computational record must explain how it was obtained. This appendix describes that relationship, while the separate technical reproduction companion retains the executable commands and implementation details.

The [project repository](#) contains the model source, analysis stages, and saved reports. The present edition rewrites the interpretation of those reports and checks several disputed claims against the implementation. It does not constitute a fresh rerun of the scientific experiments. A saved result, a source inspection, and an independently repeated experiment are different kinds of evidence.

Starting from the same object

Reproduction begins with the dataset and its transformation. The reported fly counts refer to the packed MaleCNS representation with a three-contact connection threshold. Some structural probes use a higher threshold, and the embodied fit uses six. A reader should recover the graph version and the threshold of the particular stage rather than assuming one connection count describes the entire project.

The common representation also carries cell identities, annotations, and sign conventions. The fly's stored graded proxy, the worm's transmitter convention, the circuit model's default sign table, and the embodied treatment of octopamine are not interchangeable. Repeating a sign-conditioned analysis requires the convention that actually produced it.

Positional mappings belong in the same record. An eight-bin ring, a sixteen-wedge anatomical description, and a linear fan-shaped-body column list are dif-

ferent coordinates. The earlier equivalence between one and a half bins and one 22.5-degree wedge was incorrect. Under an eight-bin full-circle convention, one and a half bins corresponds to 67.5 degrees mathematically; relating that value to anatomy requires the stated hemisphere and position mapping.

Following the computation

The structural stages measure flow, recurrence, signs, motifs, and region-specific patterns. The common-representation checks compare the fly results with the original fly analysis before extending the same operations to the worm. The operator detector then interprets selected combinations of measurements as candidate computations.

The ensemble specification records populations, parameter axes, stimuli, and perturbations. The runner instantiates the full loaded network, modifies the named pathways, and measures the selected circuit. The heading, column, and pattern probes use different observables, so their labels should not be compared as though they were one common measure of computational success.

The field calculation is a separate stage. It generates perturbation curves that can be attached to the heading report. If the heading report is generated before the field perturbation output exists, the merged comparison may be absent or stale. To refresh that comparison, generate the field perturbations first and then regenerate the heading report that incorporates them.

The benchmark and embodied evaluations form further branches. The benchmark compares estimators on synthetic shared inputs. The embodied calibration selects parameters against a multi-part objective, after which closed-loop scenarios and state manipulations examine the assembled system. A reproduction should preserve which observations were used for fitting and which were used afterward.

Replaying randomness

The experiments use seeded random generators, allowing a particular implementation and execution order to be replayed. A seed alone is not a complete specification of a stochastic experiment. Changing the order of operations can change which random values reach a given cell or stimulus.

For comparisons across models, matched input streams are more informative than merely supplying the same initial seed. For comparisons across code versions, the source revision, dependencies, effective parameters, and observation

definitions should accompany the seed. Hardware and numerical backend also belong in performance or equivalence claims.

A single replay answers whether one run can be recovered. Repeated seeds address variability. Changes to parameter sampling and classification thresholds address robustness of an ensemble conclusion. These checks are related, but none substitutes for the others.

Checking the accounting

Several corrections in this edition arose from following the denominators. The heading substrate contains 148 named cells, while eighty-eight counts only EPG plus Delta7. The calyx-reaching projection-neuron population gives a Kenyon-cell expansion ratio near 14.4, while the broader projection-neuron census gives about 5.9.

The PFN report contains thirteen shifted members but attributes the selected recurrent mechanisms only for the nine with a PFNd shift. Four PFNv-only cases remain outside that attribution. The spiking heading report's one confinement-failure member is also distinct from the field model's surviving widened bump. Preserving these distinctions prevents a summary from changing the scope of the underlying measurement.

Effective defaults require the same attention. The circuit ensemble uses adaptation, depression, and a current-based default configuration. The embodied fit uses a separate conductance-based configuration with adaptation set to zero. The source and saved parameters should be read together when reproducing a comparison between them.

What would strengthen the record

The current artifacts preserve parameter points, classified outcomes, and ranked experiments, but not every result is equally self-contained. Future reports should include all effective defaults, population selections, coordinate mappings, classification thresholds, and continuous measurements required to reconstruct a claim without inferring hidden state from the implementation.

The heading benchmark would benefit from retained per-trial paired errors, allowing uncertainty intervals to be computed directly. The field reduction needs an adjoint calculation and a small-perturbation check. The APL comparison needs alternative neuronal representations under matched stimulation. These are open scientific checks, rather than tasks already completed by the preparation of this edition.

Reproducibility makes a claim inspectable; it does not make its interpretation inevitable. The purpose of retaining the full chain is to let another reader rerun the result, challenge the observable, and test a different explanation on the same anatomical substrate.

Appendix B

References and Further Reading

The book combines established anatomical and physiological work with results from an exploratory software project. The sources below serve different roles. Some supply the reconstruction, some motivate model components, and some provide observations against which models are compared. A source used to design or calibrate a model should not also be counted as an independent test of that same choice.

Connectomes and model substrates

The male central nervous system reconstruction is distributed through the [MaleCNS project](#), a collaboration involving FlyEM at Janelia, Cambridge, the MRC Laboratory of Molecular Biology, and Google Research. The project's [download page](#) identifies the release artifacts. The counts in this book refer to the local packed and thresholded representation, rather than every object available in the source release.

Cook and colleagues' 2019 study, *Whole-animal connectomes of both *Caenorhabditis elegans* sexes*, supplies the worm comparison. The project uses the adult hermaphrodite chemical and electrical graphs distributed through Netzschleuder, with the recorded July 2020 correction. OpenWorm and WormAtlas cell tables inform the transmitter assignments, whose incomplete coverage limits the sign-conditioned analysis.

The embodied physical model comes from Vaxenburg and colleagues' flybody work, available through the [flybody repository](#). The visual front end comes from Lappalainen and colleagues' trained optic-lobe model, available through [flyvis](#). Walking and wing-motion resources used by the project are associated with [Fly-](#)

[Suite](#). These components contribute prior modelling and training information to the assembled animal.

Neural dynamics and sensory processing

Shiu and colleagues' 2024 connectome-based integrate-and-fire work in *Nature* provides part of the project's neuron-model lineage and motivates sugar, bitter, and veto probes. The current implementation contains several configurations, so attribution to that lineage should not be read as identity with every parameter or method in the published model.

Pugliese and colleagues' 2025 work on connectome simulations of walking-related central pattern generation motivates size scaling and probes of selected descending neurons. Its reported subnetwork results should be kept distinct from this project's unsuccessful full motor-mode test. Neither establishes a universal conclusion about all possible connectome-based walking models.

Turner and colleagues' 2008 study of Kenyon-cell olfactory representations in the *Journal of Neurophysiology* motivates the sparseness target. Olsen and Wilson's 2008 *Nature* study concerns lateral presynaptic inhibition in the antennal lobe. Gruntman and Turner's 2013 work concerns integration of Kenyon-cell inputs. Together these studies motivate operating regimes and candidate mechanisms; they do not directly supply every effective parameter in the simulation.

Amin, Apostolopoulou, Suárez-Grimalt, Vrontou, and Lin's 2020 paper, [Localized inhibition in the *Drosophila* mushroom body](#), is essential to interpreting the APL result. It motivates the need to compare the current point-neuron approximation with representations that preserve graded and local activity.

Heading and spatial computation

Turner-Evans and colleagues' 2020 paper, [The neuroanatomical ultrastructure and function of a biological ring attractor](#), supplies the Delta7 perturbation observation discussed in Chapter 9. That observation informed the revised inhibitory model. The quantitative width curve presented here remains a model prediction rather than a curve independently validated by citing the paper.

Hulse and colleagues' 2021 *eLife* connectome of the central complex provides anatomical context for columnar organisation and the PFN-to-h Δ B pathways. The present localised-stimulation experiments address a restricted transform hypothesis within that broader anatomical context.

Amari's 1977 work on pattern formation in lateral-inhibition neural fields provides the theoretical lineage for the continuous field. The specific phase-gain ap-

proximation used by this project still requires the adjoint-mode and numerical checks described in Chapter 8; invoking the general theory does not establish those implementation details.

Dasgupta, Stevens, and Navlakha's 2017 *Science* paper, *A neural algorithm for a fundamental computing problem*, motivates the connection between mushroom-body organisation and locality-sensitive hashing. The analogy concerns an algorithmic interpretation of expansion and sparsity. The fixed-weight, direct Kenyon-cell probe in this book does not test the full learning system.

Descending pathways and movement

Braun and colleagues' 2024 work on descending networks and Cande and colleagues' 2018 optogenetic dissection of descending control inform the association of populations with motor effects. Bidaye and colleagues' 2014 and 2020 studies concern moonwalker pathways and backward walking. Rayshubskiy and colleagues' 2020 work concerns DNa02 and steering, while Sapkal and colleagues' 2024 study concerns descending control of walking.

The escape and takeoff interpretation draws on von Reyn and colleagues' 2014 work and Namiki and colleagues' 2018 descending-neuron study. Ribeiro and colleagues' 2018 work on LC10 pathways and Deutsch and colleagues' 2020 work on pursuit and pIP10 inform the courtship-related inputs and readouts. These assignments identify experimentally motivated pathways; the supplied motor programmes remain separate components of the embodied model.

Maye and colleagues' 2007 study in *PLoS ONE* and Brembs's 2011 discussion in *Proceedings of the Royal Society B* inform the treatment of spontaneous behaviour. Geurten and colleagues' 2014 study of walking saccades and Seeds and colleagues' 2014 work on grooming sequences motivate selected behavioural statistics. Dethier's 1957 work and Kim and Dickinson's 2017 local-search study inform post-feeding behaviour. These sources motivate an externally supplied bout process, rather than demonstrating that the simulation derives those statistics from wiring.

Kim and colleagues' 2015 work on efference copy in visuomotor processing motivates the distinction between external motion and self-generated sensory input. Tammero and Dickinson's 2002 work concerns collision-avoidance saccades in flight. Dickinson's 1999 haltere work and Dickinson and colleagues' 1999 aerodynamic study inform parts of the embodied flight and reflex modelling.

Internal state and modulation

Yu and colleagues' 2016 *eLife* study of insulin, AKH, and octopaminergic signalling motivates the nutritional-state chain. Lee and Park's 2004 *Genetics* study and Yang and colleagues' 2015 *PNAS* work provide context for starvation-related activity and octopamine perturbations. The model's qualitative genotype comparison depends partly on its state-aware bout process, as Chapter 14 explains.

Suver and colleagues' 2012 *Neuron* study concerns flight-related modulation of visual processing by octopamine. Marella and colleagues' 2012 work concerns dopaminergic modulation of sucrose acceptance. These motivate state-dependent effects, but do not establish the uniform receptor action assigned to all anatomical targets in the current model.

Inagaki and colleagues' 2012 work and LeDue and colleagues' 2016 work inform hunger-dependent changes in sugar and bitter sensitivity. These sensory-state relations are additional physiological constraints, rather than quantities determined by the chemical contact graph alone.

The project reports are the sources for the simulation counts, classifications, and benchmark means in this edition. They should be read alongside these primary studies, with the distinction between a measured animal, an implemented model, and a proposed experiment kept intact.